Analyse IV

Luigi Ieracitano

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Contents

Chapter 1

Review of complex numbers $\mathbb{C}$

1.1 Basic Definitions

A complex number is written as

$$z = x + iy, \qquad x = \Re(z) \in \mathbb{R},\ y = \Im(z) \in \mathbb{R},\ i^2 = -1.$$

Conjugate: $\overline{z} = x - iy$.
Modulus:

$$|z| = \sqrt{x^2 + y^2}.$$

Geometric interpretation: $|z|$ is the distance from z to 0, and $|z - z_0|$ is the distance from z to z_0.

1.2 Euler's Formula

For $\theta \in \mathbb{R}$,

$$e^{i\theta} = \cos\theta + i\sin\theta, \quad \text{thus} \quad |e^{i\theta}| = 1.$$

1.3 Polar Coordinates

If $z \neq 0$, there exist $r = |z| > 0$ and an argument θ such that

$$z = re^{i\theta}, \qquad \theta \equiv Arg(z) \pmod{2\pi}.$$

Principal argument: $Arg(z) \in (-\pi, \pi]$.
Products/quotients in polar form: if $z_1 = r_1 e^{i\theta_1}$, $z_2 = r_2 e^{i\theta_2}$,

$$z_1 z_2 = r_1 r_2 e^{i(\theta_1+\theta_2)}, \qquad \frac{z_1}{z_2} = \frac{r_1}{r_2} e^{i(\theta_1-\theta_2)} \quad (z_2 \neq 0).$$

Rotation: multiplying by $e^{i\phi}$ corresponds to a rotation of angle ϕ.

1.4 Algebraic Equations: k-th Roots

To solve $z^k = a$ $(a \neq 0)$, write $a = \rho e^{i\alpha}$:

$$z = \rho^{1/k} e^{i(\alpha + 2\pi m)/k}, \qquad m = 0, 1, \ldots, k-1.$$

Thus there are k **solutions** (counting multiplicities).

1.5 Complex functions

A function $f : \Omega \subset \mathbb{C} \to \mathbb{C}$ can be written as

$$f(z) = f(x + iy) = u(x, y) + iv(x, y),$$

where $u, v : \Omega \subset \mathbb{R}^2 \to \mathbb{R}$.

1.5.1 Complex Exponential and Trigonometry

$$e^{x+iy} = e^x(\cos y + i \sin y).$$

Definitions (via the exponential, consistent with Euler's formula):

$$\cos z = \frac{e^{iz} + e^{-iz}}{2}, \qquad \sin z = \frac{e^{iz} - e^{-iz}}{2i}.$$

Hyperbolic functions:

$$\cosh z = \frac{e^z + e^{-z}}{2}, \qquad \sinh z = \frac{e^z - e^{-z}}{2}.$$

1.6 Continuity

f is continuous at z_0 if

$$\forall \varepsilon > 0,\ \exists \delta > 0 :\ |z - z_0| < \delta \Rightarrow |f(z) - f(z_0)| < \varepsilon.$$

Equivalent: $\lim_{z \to z_0} f(z) = f(z_0)$.
Practical criterion: if $f = u + iv$, then f is continuous at $z_0 = x_0 + iy_0$

$$\iff u \text{ and } v \text{ are continuous at } (x_0, y_0).$$

1.7 Complex Derivative

f is **complex differentiable** at z_0 if the limit exists:

$$f'(z_0) = \lim_{z \to z_0} \frac{f(z) - f(z_0)}{z - z_0}.$$

1.8 Holomorphic and Entire Functions

Let $\Omega \subset \mathbb{C}$ be **open**.

- f is **holomorphic** on Ω if it is differentiable at every point of Ω.
- f is **entire** if it is holomorphic on $\mathbb{C}$.

Recall: Ω is open if $\forall z_0 \in \Omega,\ \exists r > 0$ such that the disk $D_r(z_0) = \{z : |z - z_0| < r\} \subset \Omega$.

1.9 Cauchy–Riemann Equations

1.9.1 Theorem (C–R)

Let Ω be open and $f = u + iv$ with $u, v \in C^1(\Omega)$. If f is holomorphic, then

$$\boxed{\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \qquad \text{and} \qquad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}} \quad \text{(Cauchy–Riemann equations).}$$

1.9.2 Useful Formula for f'

When C–R are satisfied,

$$\boxed{f'(z) = u_x + iv_x = v_y - iu_y.}$$

1.9.3 Quick Examples

- $f(z) = z^2 = (x+iy)^2 = (x^2 - y^2) + i(2xy)$: $u = x^2 - y^2$, $v = 2xy$. We have $u_x = 2x = v_y$ and $u_y = -2y = -v_x$, hence f is holomorphic and $f'(z) = 2z$.
- Conjugation $f(z) = \overline{z} = x - iy$: $u = x$, $v = -y$, so $u_x = 1$ but $v_y = -1 \Rightarrow$ C–R fails: **not holomorphic**.
- $f(z) = e^z = e^x(\cos y + i \sin y)$: C–R holds $\Rightarrow$ holomorphic and $(e^z)' = e^z$.
- $f(z) = \frac{1}{z}$ is holomorphic on $\mathbb{C}^*$ and $f'(z) = -\frac{1}{z^2}$.

1.9.4 Differentiation Rules (same as real case)

If f, g are holomorphic on Ω:

$$(f+g)' = f' + g', \qquad (fg)' = f'g + fg', \qquad \left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2} \quad (g \neq 0).$$

Chain rule: if $g(\Omega) \subset W$ and f is holomorphic on W, then

$$(f \circ g)'(z) = f'(g(z))\, g'(z).$$

Usual consequences:

$$(z^k)' = kz^{k-1}, \qquad (\cos z)' = -\sin z, \qquad (\sin z)' = \cos z.$$

1.9.5 Geometric Interpretation of C–R (idea)

View $f : \mathbb{R}^2 \to \mathbb{R}^2$ via $(x, y) \mapsto (u(x, y), v(x, y))$. If f is holomorphic, the Jacobian matrix Df has the form

$$Df = \begin{pmatrix} u_x & u_y \\ v_x & v_y \end{pmatrix} = \begin{pmatrix} a & -b \\ b & a \end{pmatrix},$$

so locally it is a **scaling** (factor $\sqrt{a^2 + b^2}$) followed by a **rotation** (no anisotropic stretching): small squares remain squares.

1.10 Complex Logarithm

1.10.1 Argument and Discontinuity

$Arg(z)$ (principal value) is defined on $\mathbb{C} \setminus \{0\}$ but is **discontinuous** along the negative real axis (jump from π to $-\pi$).

$$\operatorname{Arg}(z) = \begin{cases} \arctan\left(\dfrac{y}{x}\right) & \text{if } x > 0 \\ \arctan\left(\dfrac{y}{x}\right) + \pi & \text{if } x < 0 \text{ and } y \geq 0 \\ \arctan\left(\dfrac{y}{x}\right) - \pi & \text{if } x < 0 \text{ and } y < 0 \\ \dfrac{\pi}{2} & \text{if } x = 0 \text{ and } y > 0 \\ -\dfrac{\pi}{2} & \text{if } x = 0 \text{ and } y < 0 \end{cases}$$

1.10.2 Definition of the Principal Logarithm

The principal branch of the complex logarithm is

$$\boxed{Log(z) = \log|z| + i\,Arg(z)}$$

and is defined on $\mathbb{C}^*$ and holomorphic on $\mathbb{C} \setminus \{(-\infty, 0)\}$.

1.10.3 Properties and Remarks

- On a domain where a branch is chosen (e.g. $\mathbb{C} \setminus \{x \leq 0\}$), Log is holomorphic and
$$(Logz)' = \frac{1}{z}.$$
- Usual rules do not hold *strictly* due to multivaluedness/discontinuity:
$$Log(z_1 z_2) \neq Logz_1 + Logz_2 \quad \text{in general,}$$
but they hold **up to a multiple of** $2\pi i$:
$$Log(z_1 z_2) = Logz_1 + Logz_2 + 2\pi i k, \qquad k \in \mathbb{Z}.$$

Chapter 2

Complex integration

2.1 Line Integrals in $\mathbb{R}^2$ (Recall)

Let $F : \mathbb{R}^2 \to \mathbb{R}^2$ be a continuous vector field

$$F(x_1, x_2) = (F_1(x_1, x_2), F_2(x_1, x_2))$$

Let $\gamma : [a, b] \to \mathbb{R}^2$ be a C^1 curve with image Γ.
The line integral of F along γ is defined as

$$\int_\Gamma F \cdot d\ell = \int_a^b F(\gamma(t)) \cdot \gamma'(t)\, dt$$

which explicitly gives

$$\int_a^b \left[F_1(\gamma(t))\gamma_1'(t) + F_2(\gamma(t))\gamma_2'(t)\right] dt$$

Remarks

- The line integral is independent of the parametrization.
- Reversing the orientation changes the sign.
- A curve is called
  - **simple** if it has no self-intersections
  - **regular** if $\gamma'(t) \neq 0$

If Γ is closed and $\nabla \times F = 0$ in the interior of Γ then by Stoke's Theorem

$$\int_\Gamma F \cdot d\ell = 0$$

2.2 Complex Line Integrals

Let $\gamma : [a, b] \to \mathbb{C}$ be a regular curve parametrizing $\Gamma \subset \mathbb{C}$.
Let $f : \Gamma \to \mathbb{C}$ be continuous.

Definition 1. The line integral of f along Γ is

$$\int_\Gamma f(z)\, dz = \int_a^b f(\gamma(t))\gamma'(t)\, dt$$

Orientation

If $-\Gamma$ denotes the curve with opposite orientation then

$$\int_{-\Gamma} f(z)\,dz = -\int_{\Gamma} f(z)\,dz$$

Remark

The value of the line integral does not depend on the parametrization as long as the orientation is preserved.

Examples Integral of z^2 on the unit circle

Let Γ be the unit circle parametrized by

$$\gamma(t) = e^{it}, \quad t \in [0, 2\pi]$$

Then

$$\gamma'(t) = ie^{it}$$

We compute

$$\int_{\Gamma} z^2\,dz = \int_0^{2\pi} (e^{it})^2(ie^{it})\,dt$$

$$= i\int_0^{2\pi} e^{3it}dt$$

$$= i\left[\frac{e^{3it}}{3i}\right]_0^{2\pi} = 0$$

Integral of $\frac{1}{z}$

$$\int_{\Gamma} \frac{1}{z}dz = \int_0^{2\pi} \frac{1}{e^{it}}(ie^{it})dt = i\int_0^{2\pi} dt = 2\pi i$$

Integral of $\frac{1}{z^2}$

$$\int_{\Gamma} \frac{1}{z^2}dz = \int_0^{2\pi} \frac{1}{e^{2it}}(ie^{it})dt = i\int_0^{2\pi} e^{-it}dt = 0$$

2.3 Cauchy's Theorem

Theorem 1. *Let $U \subset \mathbb{C}$ be open and simply connected.*
Let Γ be a closed piecewise regular curve contained in U.
If $f : U \to \mathbb{C}$ is holomorphic then

$$\int_\Gamma f(z)\, dz = 0$$

Simply connected: the domain has no holes.
Examples:

$$\mathbb{C}, \quad \{z \in \mathbb{C} : |z| < 1\}$$

are simply connected, while

$$\mathbb{C}^* = \mathbb{C} \setminus \{0\}$$

is not.
Piecewise regular: Γ might have finitely many curves, e.g. square.

2.3.1 Idea of the Proof

Write

$$f(z) = u(x, y) + iv(x, y)$$

and parametrize

$$\gamma(t) = \alpha(t) + i\beta(t)$$

Then

$$\int_\Gamma f(z)\, dz = \int_a^b \left[u(\gamma(t)) + iv(\gamma(t))\right] (\alpha'(t) + i\beta'(t))dt$$

Separating real and imaginary parts gives two line integrals in $\mathbb{R}^2$.
Using Green's theorem and the Cauchy–Riemann equations

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}, \qquad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

both integrals vanish, hence

$$\int_\Gamma f(z)\, dz = 0.$$

2.3.2 Extension of Cauchy's Theorem

Corollary 1. *Let Γ_0 and Γ_1 be closed simple regular curves with*

$$\Gamma_1 \subset int(\Gamma_0)$$

If f is holomorphic on the region between them then

$$\int_{\Gamma_0} f(z)\,dz = \int_{\Gamma_1} f(z)\,dz$$

This allows computation of complicated integrals using simpler curves.

2.4 Cauchy Integral Formula

Theorem 2. *Let $D \subset \mathbb{C}$ be open and $f : D \to \mathbb{C}$ holomorphic.*
Let Γ be a closed simple curve inside D oriented clockwise and let z_0 lie inside Γ.
Then

$$f(z_0) = \frac{1}{2\pi i} \int_\Gamma \frac{f(z)}{z - z_0} dz$$

This formula is particularly useful to compute integrals of the form

$$\int_\Gamma \frac{f(z)}{z - z_0}\,dz.$$

Remark: if the curves is oriented counterclockwise we change the sign as usual.
If more than one z_0, we can use this formula if only one z_0 is inside Γ, if yes use this formula by using this z_0.
E.g:

$$I = \int_\gamma \frac{e^z}{(z-1)(z^2+4)} dx \quad \gamma = e^{it} \quad t \in [\,0, 2\pi]$$

Here $z_0 = 2i, -2i$ is out of the unit circle, so only do for $z_0 = 1$

$$f(z) = \frac{e^z}{z^2+4}$$

Idea of the proof

Consider a small circle around z_0

$$\gamma_r(t) = z_0 + re^{it}$$

Then

$$\int_\Gamma \frac{f(z)}{z - z_0} dz = \int_{\gamma_r} \frac{f(z)}{z - z_0} dz$$

which becomes

$$\int_0^{2\pi} f(z_0 + re^{it}) i dt$$

As $r \to 0$, continuity of f gives

$$2\pi i f(z_0)$$

which proves the formula.

Lecture 4 — 12-03-2025: Laurent series and Taylor series

2.4.1 Extended Cauchy Integral Formula

Under the same assumptions, for every integer $n \geq 0$ we have

Corollary 2.

$$f^{(n)}(z_0) = \frac{n!}{2\pi i} \int_\Gamma \frac{f(z)}{(z - z_0)^{n+1}}\, dz.$$

Equivalently,

$$\int_\Gamma \frac{f(z)}{(z - z_0)^{n+1}}\, dz = \frac{2\pi i}{n!} f^{(n)}(z_0).$$

Remark. This formula shows that holomorphic functions are infinitely differentiable.

Examples Let $f(z) = \frac{\cos z}{z}$ and let Γ be a positively oriented simple closed curve. We distinguish three cases:

- If 0 lies on Γ, the integral is not defined.
- If 0 is outside Γ, then

$$\int_\Gamma \frac{\cos z}{z}\, dz = 0$$

  by Cauchy's theorem.
- If 0 is inside Γ, then using Cauchy's integral formula

$$\int_\Gamma \frac{\cos z}{z}\, dz = 2\pi i \cos(0) = 2\pi i.$$

Let $\Gamma = \{z \in \mathbb{C} : |z - 2| = 1\}$ oriented counterclockwise. Compute

$$\int_\Gamma \frac{e^z}{(z - 2)^3}\, dz.$$

Using the extended Cauchy formula with $f(z) = e^z$, $z_0 = 2$ and $n = 2$,

$$\int_\Gamma \frac{e^z}{(z - 2)^3}\, dz = \frac{2\pi i}{2!} f^{(2)}(2).$$

Since

$$f^{(2)}(z_0) = e^2,$$

we obtain

$$\int_\Gamma \frac{e^z}{(z - 2)^3}\, dz = \frac{2\pi i}{2} e^2 = \pi i e^2.$$

Chapter 3

Laurent series and Taylor series

3.1 Taylor Series

Let $D \subset C$ open, $z_0 \in D$, and $R > 0$ such, that the disk $\{z \in C : |z - z_0| < r\}$ is contained in D. If $f : D \to C$ is holomorphic, it has a Taylor series expansion for every $z \in C$ with $|z - z_0| < R$

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(x_0)}{n!}(x - x_0)^n.$$

As Taylor series around a point is unique, one can compute Taylor series using known series expansions. The largest possible R is called **radius of convergence** (It's the distance between z_0 and the nearest singularity (not z_0).

Example For the exponential function,

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}.$$

This series converges for all $x \in \mathbb{R}$.
Remark: Functions f s.t the Taylor series converges to f in a neighborood of x_0 are called **analytic**.

3.1.1 Taylor coefficients

Definition 2 (Taylor Coefficients)**.** The Taylor coefficients of a function $f(z)$ around a point z_0 are defined as:

$$c_n = \frac{f^{(n)}(z_0)}{n!}$$

where $f^{(n)}(z_0)$ is the n-th derivative of f evaluated at z_0.
This comes from assuming f has a power series representation:

$$f(z) = \sum_{n=0}^{\infty} c_n(z - z_0)^n$$

Remark:
(1) All holo functions are analytic.
(2) In practice: R is the distance of z_0 from the *nearest* point where f is not holo or not defined.
(3) We can connect the Taylor series to Cauchy's integral formula

$$f(z) = \sum_{n=0}^{\infty} \frac{f^{(n)}(z_0)}{n!}(z - z_0)^n = \sum_{n=0}^{\infty} c_n(z - z_0)^n$$

where

$$c_n = \frac{f^{(n)}(z_0)}{n!} = \frac{1}{2\pi i}\int_\gamma \frac{f(\zeta)}{\zeta - z_0} d\zeta$$

for every simple, close curve γ aroung z_0 with int $\gamma \subset D$ and counterclockwise oriented.

3.2 Laurent series

Theorem 3 (Laurent series)**.** *Let $D \subset \mathbb{C}$ be open and let $z_0 \in D$. Assume $f : D \setminus \{z_0\} \to \mathbb{C}$ is holomorphic.*
Let $R > 0$ such that $B_R(z_0) \subset D$.
Then for all $z \in B_R(z_0) \setminus \{z_0\}$

$$f(z) = \sum_{n=-\infty}^{\infty} c_n (z - z_0)^n$$

where for every simple closed counterclockwise oriented curve γ such that $\operatorname{int}(\gamma) \subset D$ *and z_0 is enclosed by γ,*

$$c_n = \frac{1}{2\pi i}\int_\gamma \frac{f(\zeta)}{(\zeta - z_0)^{n+1}}\, d\zeta.$$

Remark

- The Laurent series is like a Taylor series but it may include terms with polynomial singularities (negative powers).
- As for Taylor series, the Laurent series representation is unique. Hence if we determine the coefficients c_n by any method (not necessarily by path integrals), we have determined the Laurent series.
- We do not assume that f is holomorphic at z_0. It might be holomorphic there or not; no hypothesis is made.

Definition 3. Any Laurent series can be split into a **singular part** and a **regular part**:

$$\sum_{n=-\infty}^{\infty} c_n (z - z_0)^n = \underbrace{\sum_{n=1}^{\infty} c_{-n} (z - z_0)^{-n}}_{\text{singular part}} + \underbrace{\sum_{n=0}^{\infty} c_n (z - z_0)^n}_{\text{regular part}}.$$

In other words, a Taylor series is a Laurent series whose singular part vanishes.

Example If $f : D \to \mathbb{C}$ is holomorphic and $z_0 \in D$, then its Taylor expansion

$$f(z) = \sum_{n=0}^{\infty} c_n (z - z_0)^n$$

is a Laurent series whose singular part is zero since for all $n \leq -1$

$$\frac{f(\zeta)}{(\zeta - z_0)^{n+1}} \text{is holo. near} z_0$$

so by Cauchy's integral formula $\int_\gamma \frac{f(\zeta)}{(\zeta - z_0)^{n+1}} d\zeta = 0$

Example For the following function $f : \mathbb{C} \setminus 0 \to \mathbb{C}$ let find the Laurent series around $z_0 = 0$
1) $f(z) = \frac{1}{z}$, this is already a Laurent serieswith

$$c_{-1} = 1, \qquad c_n = 0 \text{ otherwise.}$$

2) $f(z) = \frac{5z+1}{z^2} = \frac{1}{z^2} + \frac{5}{z} = \sum_{n=-\infty}^{\infty} c_n z^n$ where

$$c_n = \begin{cases} 1 & \text{if } n = -2 \\ 5 & \text{if } n = -1 \\ 0 & otherwise \end{cases}$$

Lecture 5 — 19-03-2026: Singularities

Example We know:

$$\frac{1}{1-z} = \sum_{n=0}^{\infty} z^n, \quad |z| < 1$$

Then:

$$\frac{1}{1+z^2} = \frac{1}{1-(-z^2)} = \sum_{n=0}^{\infty} (-1)^n z^{2n}$$

So:

$$\frac{1}{1+z^2} = \sum_{k=0}^{\infty} c_k z^k$$

where

$$c_k = \begin{cases} 0 & \text{if } k \text{ odd} \\ (-1)^{k/2} & \text{if } k \text{ even} \end{cases}$$

Example: Laurent expansion around z_0 Let $f : \mathbb{C} \setminus \{0, -1\} \to \mathbb{C}$

$$f(z) = \frac{1}{z(z+1)}$$

We study the Laurent expansion at z_0.
Case 1: $z_0 = 0$

$$f(z) = \frac{1}{z} - \frac{1}{z+1}$$

We expand:

$$\frac{1}{z+1} = \sum_{n=0}^{\infty} (-1)^n z^n, \quad |z| < 1$$

Thus:

$$f(z) = \frac{1}{z} - \sum_{n=0}^{\infty} (-1)^n z^n = \sum_{n=-1}^{\infty} (-1)^{n+1} z^n$$

Radius of convergence:

$$R = 1$$

as the nearest singularity is at $z = -1$ (distance of z_0 to the next distinct singular point of f)
Case 2: $z_0 = -1$

$$f(z) = -\frac{1}{z+1} + \frac{1}{z}$$

Expand around $z = -1$:

$$\frac{1}{z} = \frac{1}{(z+1)-1} = \frac{1}{-1(1-(z+1))} = \frac{-1}{1-(z+1)} = \sum_{n=0}^{\infty}(-1)(z+1)^n, \quad |z+1| < 1$$

So:

$$f(z) = -\frac{1}{z-(-1)} + \sum_{n=0}^{\infty}(-1)(z+1)^n = \sum_{n=1}^{\infty}(-1)(z+1)^n$$

Radius of convergence:

$$R = 1$$

(Here the center = z_0 and the nearest singularity is at z = 0 hence the distance is $|0-(-1)| = 1$)
Case 3: $z_0 \neq 0, -1$

f is holomorphic near z_0, so f can be written as a regular Taylor series. Converges in any disk around z_0 which does not contains neither 0 and -1.
Radius:

$$R = \min\{|z_0|, |z_0+1|\}$$

3.3 Types of Singularities

3.3.1 Regular Point

Definition 4. f has a **regular point** at z_0 if the singular part of its Laurent series at z_0 vanishes.

Remark 1. f is holomorphic at $z_0 \iff z_0$ is a regular point.
z_0 is a regular point $\iff lim_{z \to z_0} f(z)$ exists

$$f(z) = e^z = \sum_{n=0}^{\infty} \frac{z^n}{n!}$$

$z_0 = 0$ is a regular point (no terms of the form $z^{-1}, z^{-2}, \dots \ (n < 0)$)

3.3.2 Pole

Definition 5. f has a **pole of order** m at z_0 if for $n < -m, c_n = 0, c_{-m} \neq 0$:

$$f(z) = \frac{c_{-m}}{(z-z_0)^m} + \dots + \frac{c_{-1}}{z-z_0} + \text{regular terms}$$

$$f(z) = \frac{1}{z} \quad \Rightarrow \text{pole of order 1 at } z = 0$$

$$f(z) = \frac{1}{(z+2)^2} + \frac{1}{z}$$

pole of order 2 at $z = -2$, pole of order 1 at $z = 0$.

Remark 2. f has a pole of order m at $z_0 \iff \lim_{z \to z_0}(z-z_0)^m f(z)$ exists and is not zero.

In particular if f has a pole of order m at z_0 we can write

$$f(z) = \frac{g(z)}{(z - z_0)^m}$$

where g is holo. and $g(z_0) \neq 0$ (opposite is also true).

3.3.3 Essential Singularity

Definition 6. f has an **essential singularity** at z_0 if its Laurent series has infinitely many coefficient $c_n \neq 0$ with $n \leq -1$.

$f(z) = e^{1/z}$ at $z_0 = 0$

$$f(z) = \sum_{n=0}^{\infty} \frac{1}{n!} z^{-n} = \sum_{n=-\infty}^{0} \frac{1}{(-n)!} z^n$$

Essential singularity at $z = 0$.

3.4 Quotients of Taylor Series

Let $p, q : D \setminus \{z_0\} \to \mathbb{C}$ holo with z_0 being regular for both. Write their Laurent series as:

$$p(z) = a_0 + a_1(z - z_0) + \cdots + a_k(z - z_0)^k \quad q(z) = b_0 + b_1(z - z_0) + \cdots + b_\ell(z - z_0)^\ell$$

Consider:

$$f(z) = \frac{p(z)}{q(z)}$$

if $b_0 \neq 0$ so $q(z_0) \neq 0$. We have

$$\lim_{z \to z_0} \frac{p(z)}{q(z)} = \frac{p(z_0)}{q(z_0)} = \frac{a_0}{b_0} \quad \text{exists, so } z_0 \text{ is a regular point for } \frac{p}{q}$$

if $a_0 = b_0 = 0$ but $b_1 \neq 0$ then we have

$$\lim_{z \to z_0} \frac{a_1(z - z_0) + a_2(z - z_0)^2 + \dots}{b_1(z - z_0) + b_2(z - z_0)^2 + \dots} = \lim_{z \to z_0} \frac{a_1 + a_2(z - z_0) + \dots}{b_1 + b_2(z - z_0) + \dots} = \frac{a_1}{b_1}$$

exists hence z_0 is a regular point for f. (similarly if $a_0 = b_0 = 0$ and $a_1 = b_1 = 0$ yet $b_2 \neq 0$)
In summary, either expand in power series and take out as many common powers of $z - z_0$ from the nominator and denominator, and take the **limit** or use **l'Hôpital's rule**:

- If $k \geq \ell$: regular point, as the limit $= \begin{cases} 0 & \text{if } k > \ell \\ \frac{a_k}{b_\ell} & \text{if } k = \ell \end{cases}$ exists
- If $k < \ell$: pole of order $\ell - k$

Example

$$f(z) = \frac{z}{\sin z}$$

where $z_0 = 0$

Using:

$$\sin z = z - \frac{z^3}{3!} + \cdots$$

We get:

$$f(z) = \frac{z}{z(1 - z^2/3! + \cdots)} = \frac{1}{1 - z^2/6 + \cdots}$$

Hence $\lim_{z \to z_0} = 1$ exist, so $z = 0$ is a regular point.

Example

$$f(z) = \frac{\sin z}{2z^3 + z^4}$$

Using expansions:

$$\sin z = z - \frac{z^3}{6} + \cdots$$

We get:

$$f(z) = \frac{1}{z^2} \cdot (\text{holomorphic nonzero function})$$

So pole of order 2 at $z = 0$.

Lecture 6 — 26-03-2026: Residues and Residue Theorem

3.5 Residues

Definition 7 (Residues)**.** Let f be holomorphic on a punctured neighborhood of z_0, and write its Laurent series at z_0 as

$$f(z) = \sum_{n=-\infty}^{\infty} c_n (z - z_0)^n.$$

The coefficient c_{-1} of the term $(z - z_0)^{-1}$ is called the *residue* of f at z_0:

$$\mathrm{Res}_{z_0}(f) = c_{-1}$$

3.5.1 Integral expression of the residue

By the definition of the coefficients in the Laurent series,

$$\mathrm{Res}_{z_0}(f) = c_{-1} = \frac{1}{2\pi i} \int_\gamma f(z)\, dz,$$

where γ is a closed, simple, counterclockwise oriented curve that contains z_0 in its interior, and such that f is holomorphic on $int_\gamma \setminus \{z_0\}$.

Example If f is holomorphic on an open set containing z_0, then

$$\mathrm{Res}_{z_0}(f) = 0$$

by Cauchy's theorem.
For

$$f(z) = \frac{1}{z},$$

since it's already the Laurent series at 0, hence

$$\mathrm{Res}_0\left(\frac{1}{z}\right) = 1$$

3.5.2 Formula for a pole of order m

Suppose that f has a pole of order m at z_0, so that its Laurent series is

$$f(z) = c_{-m}(z - z_0)^{-m} + \cdots + c_{-1}(z - z_0)^{-1} + c_0 + c_1(z - z_0) + \cdots$$

Therefore,

Corollary 3.

$$c_{-1} = \lim_{z \to z_0} \frac{1}{(m-1)!} \frac{d^{m-1}}{dz^{m-1}}\Big((z - z_0)^m f(z)\Big)$$

This formula is useful when we already know that f has a pole of order m at z_0.
In particular, if $m = 1$, then

$$\mathrm{Res}_{z_0}(f) = \lim_{z \to z_0} (z - z_0) f(z).$$

Example

$$f(z) = \frac{1}{\sin z}$$

At $z_0 = 0$, this function has a pole of order 1.
Hence,

$$\mathrm{Res}_0(f) = \lim_{z \to 0} z\, f(z) = \lim_{z \to 0} \frac{z}{\sin z}$$

Using

$$\sin z = z - \frac{z^3}{3!} + \cdots,$$

we get

$$\frac{z}{\sin z} = \frac{z}{z - \frac{z^3}{3!} + \cdots} = \frac{1}{1 - \frac{z^2}{3!} + \cdots},$$

so the limit is

$$\mathrm{Res}_0(f) = 1.$$

3.5.3 Formula for quotient

Corollary 4 (Residue at a simple pole of a quotient)**.** *Let p, q be holomorphic in a neighborhood of z_0. If*

$$q(z_0) = 0 \qquad \text{and} \qquad q'(z_0) \neq 0,$$

then $f = \dfrac{p}{q}$ has a simple pole at z_0 and

$$Res_{z_0}\left(\frac{p}{q}\right) = \frac{p(z_0)}{q'(z_0)}.$$

Remark 3. Indeed, since z_0 is a simple zero of q, we can write $q(z) = (z - z_0)\,\tilde{q}(z)$ with $\tilde{q}(z_0) = q'(z_0) \neq 0$. Hence

$$Res_{z_0}\left(\frac{p}{q}\right) = \lim_{z \to z_0} (z - z_0)\,\frac{p(z)}{q(z)} = \lim_{z \to z_0} \frac{p(z)}{\tilde{q}(z)} = \frac{p(z_0)}{q'(z_0)}.$$

3.6 Residue theorem

Theorem 4 (Residue theorem)**.** *Suppose $D \subseteq \mathbb{C}$ is open and simply connected, and let γ be a simple closed curve in D which is counterclockwise oriented. Let $z_1, \ldots, z_m \in \operatorname{int}(\gamma)$. Assume*

$$f : D \setminus \{z_1, \ldots, z_m\} \to \mathbb{C}$$

is holomorphic. Then

$$\int_\gamma f(z)\,dz = 2\pi i \sum_{i=1}^{m} Res_{z_i}(f)$$

Hence, in order to compute $\int_\gamma f(z)\,dz$, it suffices to compute the residues of the singular points in the interior of γ.

3.6.1 Idea of the proof

For simplicity, assume there are only two singular points z_1 and z_2. One deforms the contour γ into two small positively oriented curves γ_1 and γ_2 around the singularities, connected by a very narrow bridge. By Cauchy's theorem, the integral over γ is equal to the sum of the integrals over the small circles, since the contributions of the bridge cancel. Therefore,

$$\int_\gamma f(z)\,dz = \int_{\gamma_1} f(z)\,dz + \int_{\gamma_2} f(z)\,dz = 2\pi i\ \mathrm{Res}_{z_1}(f) + 2\pi i\ \mathrm{Res}_{z_2}(f).$$

Example

$$f(z) = \frac{1}{z}.$$

Let γ be a simple closed counterclockwise oriented curve.

1. If $0 \in \operatorname{int}(\gamma)$, then by the residue theorem,

$$\int_\gamma \frac{1}{z}\, dz = 2\pi i \operatorname{Res}_0\left(\frac{1}{z}\right) = 2\pi i.$$

2. If $0 \notin \operatorname{int}(\gamma)$, then

$$\int_\gamma \frac{1}{z}\, dz = 0.$$

3. If $0 \in \gamma$, then the integral is not well-defined.

Example

$$f(z) = \frac{1}{z(z+1)}.$$

We have two simple poles, at $z = 0$ and $z = -1$.
Their residues are

$$\operatorname{Res}_0(f) = \lim_{z\to 0} z f(z) = \lim_{z\to 0} \frac{1}{z+1} = 1,$$

and

$$\operatorname{Res}_{-1}(f) = \lim_{z\to -1} (z+1) f(z) = \lim_{z\to -1} \frac{1}{z} = -1.$$

So:

1. If neither 0 nor -1 lies in $\operatorname{int}(\gamma)$, then

$$\int_\gamma f(z)\, dz = 0.$$

2. If $0 \in \operatorname{int}(\gamma)$ but $-1 \notin \operatorname{int}(\gamma)$, then

$$\int_\gamma f(z)\, dz = 2\pi i.$$

3. If $-1 \in \operatorname{int}(\gamma)$ but $0 \notin \operatorname{int}(\gamma)$, then

$$\int_\gamma f(z)\, dz = -2\pi i.$$

4. If both 0 and -1 lie in $\operatorname{int}(\gamma)$, then

$$\int_\gamma f(z)\, dz = 2\pi i\big(\operatorname{Res}_0(f) + \operatorname{Res}_{-1}(f)\big) = 2\pi i(1-1) = 0.$$

5. If $0 \in \gamma$ or $-1 \in \gamma$, then the integral is not well-defined.

Example $f : \mathbb{C} \setminus \{0,1\} \to \mathbb{C}$,

$$f(z) = \frac{1}{z^2} + \frac{2}{z} + \frac{3}{z-1}.$$

Then f has a pole of order 2 at 0 and a pole of order 1 at 1.
At $z = 1$, since the pole is simple,

$$\mathrm{Res}_1(f) = \lim_{z\to 1}(z-1)f(z) = \lim_{z\to 1}\left(\frac{z-1}{z^2} + \frac{2(z-1)}{z} + 3\right) = 3.$$

At $z = 0$, we use the formula for a pole of order 2:

$$\mathrm{Res}_0(f) = \lim_{z\to 0}\frac{1}{1!}\frac{d}{dz}\big(z^2 f(z)\big).$$

Now

$$z^2 f(z) = 1 + 2z + \frac{3z^2}{z-1},$$

so

$$\frac{d}{dz}\big(z^2 f(z)\big) = 2 + \frac{6z}{z-1} - \frac{3z^2}{(z-1)^2}.$$

Evaluating at $z = 0$ gives

$$\mathrm{Res}_0(f) = 2.$$

Actually this could have been seen by reading directly $f(z)$.
If $\gamma(t) = 10e^{it}$ for $t \in [0, 2\pi]$, then both poles lie inside γ, hence

$$\int_\gamma f(z)\,dz = 2\pi i\big(\mathrm{Res}_0(f) + \mathrm{Res}_1(f)\big) = 2\pi i(2+3) = 10\pi i.$$

3.6.2 Useful remark

It is often advisable to first rewrite the function in a form where the residue can be read off immediately.
For example:

$$e^{1/z} = 1 + \frac{1}{z} + \frac{1}{2!}\frac{1}{z^2} + \cdots \quad \Longrightarrow \quad \mathrm{Res}_0\left(e^{1/z}\right) = 1.$$

On the other hand,

$$e^{1/z^2} = 1 + \frac{1}{z^2} + \frac{1}{2!}\frac{1}{z^4} + \cdots \quad \Longrightarrow \quad \mathrm{Res}_0\left(e^{1/z^2}\right) = 0.$$

3.7 Applications of Residues to Real Integrals

The residue theorem can be used to compute integrals over simple closed curves, and sometimes even to compute *real* integrals by replacing them with complex integrals. The strategy is always the same:

- Extend the real integrand to a complex function $f(z)$.
- Choose a closed contour γ that includes the real axis as part of its boundary.
- Apply the residue theorem to compute $\int_\gamma f(z)\,dz$.
- Show that the contribution of the non-real part of the contour vanishes in the limit.
- Recover the desired real integral.

3.7.1 A motivating example

Let us compute

$$\int_{-\infty}^{\infty} \frac{1}{1+x^2}\,dx.$$

We already know from real analysis that this integral equals π (since arctan is a primitive of $1/(1+x^2)$). Let us recover this result using complex analysis.

Set

$$f(z) = \frac{1}{1+z^2}.$$

Given $R > 0$, consider:

- the segment $L_R : [-R, R] \to \mathbb{C}$, $L_R(t) = t$;
- the upper semicircle $C_R : [0, \pi] \to \mathbb{C}$, $C_R(t) = Re^{it}$.

Let $\gamma_R = L_R \cup C_R$ be the closed counterclockwise oriented contour.

The function f has poles of order 1 at $z = \pm i$. Only $z = i$ lies in the upper half-plane, so it is the only pole inside γ_R (provided $R > 1$). By the residue theorem,

$$\int_{\gamma_R} f(z)\,dz = 2\pi i \ \mathrm{Res}_i(f).$$

The residue is computed via

$$\mathrm{Res}_i(f) = \lim_{z\to i}(z-i)f(z) = \lim_{z\to i} \frac{1}{z+i} = \frac{1}{2i}.$$

Hence

$$\int_{\gamma_R} f(z)\,dz = 2\pi i \cdot \frac{1}{2i} = \pi.$$

On the other hand, splitting the contour:

$$\int_{\gamma_R} f(z)\,dz = \int_{L_R} f(z)\,dz + \int_{C_R} f(z)\,dz = \int_{-R}^{R} \frac{1}{1+t^2}\,dt + \int_{C_R} f(z)\,dz.$$

It remains to show that the contribution of C_R vanishes as $R \to \infty$. We estimate:

$$\left|\int_{C_R} f(z)\,dz\right| = \left|\int_0^{\pi} \frac{1}{1+R^2e^{2it}}\, iRe^{it}\,dt\right| \le \int_0^{\pi} \frac{R}{R^2-1}\,dt = \frac{\pi R}{R^2-1} \xrightarrow[R\to\infty]{} 0.$$

Therefore,

$$\int_{-\infty}^{\infty} \frac{1}{1+x^2}\,dx = \pi.$$

Remark 4. This method works for any integrand that decays *strictly faster* than $1/R$ as $R \to \infty$, which ensures the integral over the arc vanishes.

3.7.2 Integrals of rational functions times e^{iax}

A more interesting class of integrals concerns functions of the form $R(x)\cos(ax)$ or $R(x)\sin(ax)$, where R is a rational function. These appear naturally in Fourier analysis, in the study of resonances in damped systems, etc.

Theorem 5 (Real integrals via residues). *Let* $R(x) = \dfrac{P(x)}{Q(x)}$ *where* P, Q *are real polynomials such that:*

1. $\deg Q \geq \deg P + 2$*;*
2. $Q(x) \neq 0$ *for all* $x \in \mathbb{R}$.

Let $a \geq 0$ *and define* $f(z) = R(z)e^{iaz}$. *Then*

$$\int_{-\infty}^{\infty} R(x)e^{iax}\,dx = 2\pi i \sum_k \operatorname{Res}_{z_k}(f),$$

where the sum is taken over the poles z_k *of* f *in the* upper half-plane $\{\Im z > 0\}$.

Observations

- The condition $\deg Q \geq \deg P + 2$ forces $\deg Q$ to be even (otherwise Q would have a real root).
- Since Q has real coefficients, its roots come in pairs of complex conjugates. None of them are real by assumption, so half of them lie in the upper half-plane and the other half in the lower half-plane.
- Using Euler's formula $e^{iax} = \cos(ax) + i\sin(ax)$, the real and imaginary parts of the integral give

$$\int_{-\infty}^{\infty} R(x)\cos(ax)\,dx = \Re\left(\int_{-\infty}^{\infty} R(x)e^{iax}\,dx\right),$$

$$\int_{-\infty}^{\infty} R(x)\sin(ax)\,dx = \Im\left(\int_{-\infty}^{\infty} R(x)e^{iax}\,dx\right).$$

Why the contour integral over the arc vanishes

The key estimate is the following: for $z = x + iy$ with $y \geq 0$ and $a \geq 0$,

$$|e^{iaz}| = |e^{ia(x+iy)}| = |e^{iax}| \cdot |e^{-ay}| = e^{-ay} \leq 1.$$

Remark 5. Note that $|e^{iaz}|$ is **not** bounded on all of $\mathbb{C}$: in the lower half-plane ($y < 0$), the term e^{-ay} blows up. This is precisely why we must close the contour in the *upper* half-plane when $a \geq 0$. For $a < 0$, one would close the contour in the lower half-plane instead.

For $z = Re^{i\theta}$ on C_R with $\theta \in [0, \pi]$, we have $\Im z = R\sin\theta \geq 0$, hence

$$|e^{iaRe^{i\theta}}| \leq 1.$$

For the rational part, write

$$P(z) = b_m z^m + \cdots, \qquad Q(z) = c_n z^n + \cdots, \qquad n \geq m + 2.$$

Then for large $|z|$,

$$|R(z)| = \left|\frac{P(z)}{Q(z)}\right| \le \frac{C}{|z|^{n-m}} \le \frac{C}{|z|^2}.$$

Therefore

$$\left|\int_{C_R} f(z)\,dz\right| \le \int_0^{\pi} |R(Re^{i\theta})| \cdot |e^{iaRe^{i\theta}}| \cdot R\,d\theta \le \int_0^{\pi} \frac{C}{R^{n-m}} \cdot 1 \cdot R\,d\theta = \frac{\pi C}{R^{n-m-1}} \xrightarrow[R\to\infty]{} 0.$$

Benefit We compute the residues of f by finding its poles and their orders, without ever computing a Laurent series explicitly.

3.7.3 A worked example

We compute

$$I = \int_{-\infty}^{\infty} \frac{x^2}{16+x^4} \cos x\,dx.$$

This fits the previous setup with $a = 1$, $P(x) = x^2$, $Q(x) = 16 + x^4$. Note $\deg Q = 4 \ge \deg P + 2 = 4$, and $16 + x^4 > 0$ for all real x. By Euler's formula,

$$I = \Re\left(\int_{-\infty}^{\infty} \frac{x^2}{16+x^4} e^{ix}\,dx\right).$$

Define

$$f(z) = \frac{z^2}{16+z^4} e^{iz}.$$

Finding the poles. The poles of f are the roots of $16 + z^4 = 0$:

$$z^4 = -16 = 16e^{i(\pi+2k\pi)}, \qquad k \in \{0, 1, 2, 3\}.$$

Therefore

$$z_k = 2e^{i\left(\frac{\pi}{4}+\frac{k\pi}{2}\right)}, \qquad k = 0, 1, 2, 3.$$

Explicitly:

$$z_1 = \sqrt{2} + i\sqrt{2}, \quad z_2 = -\sqrt{2} + i\sqrt{2}, \quad z_3 = -\sqrt{2} - i\sqrt{2}, \quad z_4 = \sqrt{2} - i\sqrt{2}.$$

Only z_1 and z_2 lie in the upper half-plane, hence only these contribute.

Computing the residues. Each z_k is a simple pole. Using the standard trick: if $g(z) = h(z)/Q(z)$ where Q has a simple zero at z_k, then

$$\mathrm{Res}_{z_k}(g) = \frac{h(z_k)}{Q'(z_k)}.$$

Here $h(z) = z^2 e^{iz}$ and $Q'(z) = 4z^3$, so:

$$\mathrm{Res}_{z_k}(f) = \frac{z_k^2 e^{iz_k}}{4z_k^3} = \frac{e^{iz_k}}{4z_k}.$$

For $z_1 = \sqrt{2} + i\sqrt{2} = 2e^{i\pi/4}$:

$$\frac{1}{4z_1} = \frac{1}{8e^{i\pi/4}} = \frac{e^{-i\pi/4}}{8} = \frac{1-i}{8\sqrt{2}}.$$

Also $e^{iz_1} = e^{i(\sqrt{2}+i\sqrt{2})} = e^{-\sqrt{2}}e^{i\sqrt{2}}$. Therefore

$$\mathrm{Res}_{z_1}(f) = \frac{e^{-\sqrt{2}}e^{i\sqrt{2}}(1-i)}{8\sqrt{2}}.$$

For $z_2 = -\sqrt{2} + i\sqrt{2} = 2e^{i3\pi/4}$:

$$\frac{1}{4z_2} = \frac{e^{-i3\pi/4}}{8} = \frac{-1-i}{8\sqrt{2}}.$$

Also $e^{iz_2} = e^{-\sqrt{2}}e^{-i\sqrt{2}}$. Therefore

$$\operatorname{Res}_{z_2}(f) = \frac{e^{-\sqrt{2}}e^{-i\sqrt{2}}(-1-i)}{8\sqrt{2}}.$$

Summing.

$$\operatorname{Res}_{z_1}(f) + \operatorname{Res}_{z_2}(f) = \frac{e^{-\sqrt{2}}}{8\sqrt{2}} \left[e^{i\sqrt{2}}(1-i) + e^{-i\sqrt{2}}(-1-i) \right].$$

Expanding using $e^{\pm i\sqrt{2}} = \cos\sqrt{2} \pm i \sin\sqrt{2}$ and collecting terms:

$$e^{i\sqrt{2}}(1-i) + e^{-i\sqrt{2}}(-1-i) = 2i\sin\sqrt{2} - 2i\cos\sqrt{2} = 2i(\sin\sqrt{2} - \cos\sqrt{2}).$$

Hence

$$\operatorname{Res}_{z_1}(f) + \operatorname{Res}_{z_2}(f) = \frac{e^{-\sqrt{2}}}{8\sqrt{2}} \cdot 2i(\sin\sqrt{2} - \cos\sqrt{2}) = \frac{i\,e^{-\sqrt{2}}}{4\sqrt{2}}(\sin\sqrt{2} - \cos\sqrt{2}).$$

Final result. Multiplying by $2\pi i$:

$$2\pi i\big(\operatorname{Res}_{z_1}(f) + \operatorname{Res}_{z_2}(f)\big) = 2\pi i \cdot \frac{i\,e^{-\sqrt{2}}}{4\sqrt{2}}(\sin\sqrt{2} - \cos\sqrt{2}) = \frac{-\pi e^{-\sqrt{2}}}{2\sqrt{2}}(\sin\sqrt{2} - \cos\sqrt{2}).$$

Taking the real part (the result is already real):

$$\boxed{\int_{-\infty}^{\infty} \frac{x^2}{16+x^4}\cos x\,dx = \frac{\pi e^{-\sqrt{2}}}{2\sqrt{2}}\big(\cos\sqrt{2} - \sin\sqrt{2}\big) = \frac{\pi\sqrt{2}}{4}\,e^{-\sqrt{2}}\big(\cos\sqrt{2} - \sin\sqrt{2}\big).}$$

3.8 Integrals of rational functions of trigonometric functions

A second important class of real integrals that can be computed via residues are integrals of the form

$$\int_0^{2\pi} \frac{P(\cos t, \sin t)}{Q(\cos t, \sin t)}\,dt,$$

where P, Q are polynomials in two variables and $Q(x,y) \neq 0$ on the unit circle $\{x^2+y^2=1\}$.

3.8.1 The substitution $z = e^{it}$

The idea is to convert this real integral over $[0, 2\pi]$ into a complex contour integral over the unit circle. Recall:

$$\cos t = \frac{e^{it} + e^{-it}}{2}, \qquad \sin t = \frac{e^{it} - e^{-it}}{2i}.$$

If we set $z = e^{it}$ then $dz = ie^{it}\,dt$, so $dt = \dfrac{dz}{iz}$, and:

$$\cos t = \frac{1}{2}\left(z + \frac{1}{z}\right), \qquad \sin t = \frac{1}{2i}\left(z - \frac{1}{z}\right).$$

As t runs from 0 to 2π, z traces the unit circle γ once counterclockwise. Hence

$$\int_0^{2\pi} \frac{P(\cos t, \sin t)}{Q(\cos t, \sin t)}\,dt = \int_\gamma f(z)\,dz,$$

where

$$f(z) = \frac{P\left(\frac{z+1/z}{2}, \frac{z-1/z}{2i}\right)}{Q\left(\frac{z+1/z}{2}, \frac{z-1/z}{2i}\right)} \cdot \frac{1}{iz}.$$

Definition 8 (Meromorphic function)**.** A function which is holomorphic on an open set D except at isolated singular points (poles) is called *meromorphic* on D.

Since P and Q are polynomials, f is meromorphic on $\mathbb{C}$. If $z_1, \ldots, z_\ell$ are the poles of f inside the unit disk, the residue theorem gives:

$$\int_0^{2\pi} \frac{P(\cos t, \sin t)}{Q(\cos t, \sin t)}\, dt = 2\pi i \sum_{k=1}^{\ell} \mathrm{Res}_{z_k}(f).$$

3.8.2 Example

Compute

$$J = \int_0^{2\pi} \frac{1}{2 + \cos t}\, dt.$$

This fits the framework with $P(x, y) = 1$ and $Q(x, y) = 2 + x$. Substituting $z = e^{it}$:

$$2 + \cos t = 2 + \frac{1}{2}\left(z + \frac{1}{z}\right) = \frac{z^2 + 4z + 1}{2z}.$$

Therefore

$$f(z) = \frac{1}{\frac{z^2+4z+1}{2z}} \cdot \frac{1}{iz} = \frac{2z}{z^2 + 4z + 1} \cdot \frac{1}{iz} = \frac{2}{i} \cdot \frac{1}{z^2 + 4z + 1}.$$

Finding the poles. We solve $z^2 + 4z + 1 = 0$:

$$z = \frac{-4 \pm \sqrt{16 - 4}}{2} = -2 \pm \sqrt{3}.$$

So the poles are $z_1 = -2 + \sqrt{3} \approx -0.27$ and $z_2 = -2 - \sqrt{3} \approx -3.73$. Only z_1 lies inside the unit circle (since $|z_1| < 1$ but $|z_2| > 1$).

Computing the residue. z_1 is a simple pole, so:

$$\mathrm{Res}_{z_1}(f) = \lim_{z \to z_1} (z - z_1) f(z) = \frac{2}{i} \cdot \frac{1}{z_1 - z_2} = \frac{2}{i} \cdot \frac{1}{2\sqrt{3}} = \frac{1}{i\sqrt{3}}.$$

Conclusion.

$$J = 2\pi i \cdot \frac{1}{i\sqrt{3}} = \frac{2\pi}{\sqrt{3}} = \frac{2\pi\sqrt{3}}{3}.$$

3.9 General strategy

To summarize, the global strategy for applying the residue theorem to real integrals consists in:

1. **Identifying the type of integral** (rational function on $\mathbb{R}$, rational function times e^{iax}, trigonometric integral, ...).
2. **Extending the integrand** to a complex function $f(z)$.
3. **Choosing an appropriate closed contour** (semicircle in the upper half-plane, unit circle, ...).
4. **Showing that the auxiliary part of the contour gives a vanishing contribution** in the appropriate limit (using estimates on $|f|$).
5. **Identifying the poles** of f inside the contour and computing their residues.
6. **Applying the residue theorem** and recovering the desired real integral by taking real or imaginary parts if necessary.

Remark 6. The choice of contour depends on the integrand:

- For $\int_{-\infty}^{\infty} R(x)e^{iax}\,dx$ with $a > 0$: close in the upper half-plane.
- For $a < 0$: close in the lower half-plane (the orientation is then clockwise, hence a sign change).
- For trigonometric integrals over $[0, 2\pi]$: use the unit circle.
- For other situations (integrals over $[0, \infty)$, branch cuts, etc.) one may need keyhole contours, sectors, or other shapes — but the principle remains the same.

Chapter 4

Fourier Analysis

4.1 Review of the Fourier Transform

Definition 9 (Fourier Transform)**.** The **Fourier transform** $\hat{f} : \mathbb{R} \to \mathbb{C}$ of $f : \mathbb{R} \to \mathbb{C}$, whenever well-defined, is

$$\hat{f}(a) := \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-iax} f(x)\, dx.$$

Sometimes written $\mathcal{F}[f]$ instead of $\hat{f}$. It is well-defined if f is piecewise continuous and $\int_{-\infty}^{\infty} |f(x)|\, dx < \infty$ (symbolically, $f \in L^1(\mathbb{R}; \mathbb{C})$).

Definition 10 (Inverse Fourier Transform)**.** The **inverse Fourier transform** $\check{f} : \mathbb{R} \to \mathbb{C}$ of $\hat{f} : \mathbb{R} \to \mathbb{C}$, whenever well-defined, is

$$\check{f}(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{iax} f(a)\, da.$$

Sometimes written $\mathcal{F}^{-1}[\hat{f}]$ instead of $\check{f}$. Under appropriate hypotheses on f,

$$\mathcal{F}^{-1}[\mathcal{F}[f]] = f.$$

Remark 7. In general $\hat{f}$ is complex-valued even if f is real-valued. However, if f is real-valued and even (i.e. $f(x) = f(-x)$ for all $x \in \mathbb{R}$), then $\hat{f}$ is also real-valued.
Intuitively, one can think of f as a signal in the *physical domain* and $\hat{f}$ as its representation in the *frequency domain*: $\hat{f}(a)$ encodes "how much of frequency a is present in the signal f".

4.2 Properties of the Fourier Transform

(1) Linearity

$$\mathcal{F}[\lambda f + \mu g] = \lambda\, \mathcal{F}[f] + \mu\, \mathcal{F}[g]$$

whenever $\lambda, \mu \in \mathbb{R}$.

(2) Time Shift Let $g(x) = f(x - x_0)$. Then

$$\hat{g}(a) = e^{-iax_0} \hat{f}(a).$$

Computation.

$$\hat{g}(a) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-iax} f(x - x_0)\, dx \overset{y=x-x_0}{=} \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-ia(y+x_0)} f(y)\, dy = e^{-iax_0} \hat{f}(a).$$

□

(3) Frequency Shift Let $g(x) = e^{ia_0 x} f(x)$ for $a_0 \in \mathbb{R}$. Then

$$\hat{g}(a) = \hat{f}(a - a_0).$$

This is the "inversion" of property (2), as expected.

(4) Rescaling / Dilation Let $g(x) = f(\lambda x)$ where $\lambda \in \mathbb{R} \setminus \{0\}$. Then

$$\hat{g}(a) = \frac{1}{|\lambda|} \hat{f}\left(\frac{a}{\lambda}\right).$$

Computation. Set $y = \lambda x$ (so $dy = \lambda\, dx$). Two cases arise:

- If $\lambda > 0$: $\int_{-\infty}^{\infty} \cdots \frac{1}{\lambda}\, dy$.
- If $\lambda < 0$: $\int_{\infty}^{-\infty} \cdots \frac{1}{\lambda}\, dy = -\int_{-\infty}^{\infty} \cdots \frac{1}{\lambda}\, dy = \int_{-\infty}^{\infty} \cdots \frac{1}{|\lambda|}\, dy$.

In both cases,

$$\hat{g}(a) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-ia\frac{y}{\lambda}} f(y) \frac{dy}{|\lambda|} = \frac{1}{|\lambda|} \hat{f}\left(\frac{a}{\lambda}\right).$$

□

Remark 8 (Uncertainty Principle)**.** Pictorially, as $\lambda \to \infty$ the graph of $f(\lambda x)$ becomes very concentrated (width $\sim \frac{1}{\lambda}$), while $\frac{1}{|\lambda|}\hat{f}(a/\lambda)$ becomes very spread out. Hence, when f is "very concentrated", $\hat{f}$ is very spread out and vice versa. This is a manifestation of the **uncertainty principle**.

(5) Derivatives of f Let $g(x) = f'(x)$. Then

$$\hat{g}(a) = ia\, \hat{f}(a).$$

More generally, for all $n \in \mathbb{N}$,

$$\mathcal{F}[f^{(n)}](a) = (ia)^n\, \mathcal{F}[f](a).$$

So the Fourier transform converts differential equations with constant coefficients into *algebraic* equations on the frequency domain.

(6) Derivatives of $\hat{f}$ This is the dual property of (5): instead of looking at how the derivative of f affects $\hat{f}$, we look at how the derivative of $\hat{f}$ relates back to f. Suppose f is piecewise continuous and that both

$$\int_{-\infty}^{\infty} |f(x)|\,dx < \infty \qquad \text{and} \qquad \int_{-\infty}^{\infty} |x \cdot f(x)|\,dx < \infty.$$

Then $\hat{f}$ is differentiable and

$$\boxed{\big[\mathcal{F}[f](a)\big]' = -i\,\mathcal{F}[xf(x)](a).}$$

More generally, if $|x^k f(x)|$ is integrable on $\mathbb{R}$ for all $k \leq n$, then

$$\frac{d^n \hat{f}}{da^n}(a) = (-i)^n\,\mathcal{F}[x^n f(x)](a).$$

Computation. Differentiating under the integral sign,

$$\frac{d}{da}\hat{f}(a) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \frac{\partial}{\partial a}\big(e^{-iax}\big)\,f(x)\,dx = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} (-ix)\,e^{-iax} f(x)\,dx = -i\,\mathcal{F}[xf(x)](a).$$

The integrability of $|xf(x)|$ is what justifies differentiating under the integral. □

Remark 9. Note the duality with property (5):
- Differentiating in the *physical domain* $\Longleftrightarrow$ multiplying by ia in the *frequency domain.*
- Multiplying by $-ix$ in the *physical domain* $\Longleftrightarrow$ differentiating in the *frequency domain.*

This symmetry reflects the fact that x and a are conjugate variables under the Fourier transform. The same kind of duality reappears for the Laplace transform: $F'(z) = -\mathcal{L}[tf(t)]$.

(7) Convolution Given $f, g : \mathbb{R} \to \mathbb{C}$, the **convolution** $f * g : \mathbb{R} \to \mathbb{C}$, whenever well-defined, is

$$(f * g)(x) = \int_{-\infty}^{\infty} f(y)\,g(x-y)\,dy.$$

Some properties of convolution:
- $f * g = g * f$ (symmetry)
- $f * (g * h) = (f * g) * h$ (associativity)
- $f * (g + h) = f * g + f * h$ (distributivity)
- $(f * g)' = f' * g = f * g'$, so $f * g$ is at least as smooth as the smoother function among f and g.

The convolution can be interpreted as the "local average" of f with weight g.

Example If $g(x) = \frac{1}{2\varepsilon}\mathbf{1}_{[-\varepsilon,\varepsilon]}(x)$, then

$$(f * g)(x) = \int_{-\infty}^{\infty} f(y)g(x-y)\,dy = \frac{1}{2\varepsilon}\int_{x-\varepsilon}^{x+\varepsilon} f(y)\,dy,$$

By putting $x - y = -\varepsilon$ and $x - y = \varepsilon$.

which is the local average of f around x. As $\varepsilon \to 0$, $(f * g)(x) \to f(x)$ whenever f is continuous at x.

The Fourier transform converts convolutions into products:

$$\mathcal{F}[f * g] = \sqrt{2\pi}\,\mathcal{F}[f] \cdot \mathcal{F}[g].$$

In applications, one can often express $\mathcal{F}[f]$ as a product of known Fourier transforms, which allows one to recover f by convolution.

(8) Gaussian Function If $f(x) = e^{-x^2/2}$, then

$$\hat{f}(a) = e^{-a^2/2}.$$

(9) Plancherel's Theorem

$$\int_{-\infty}^{\infty} |f(x)|^2 \, dx = \int_{-\infty}^{\infty} |\hat{f}(a)|^2 \, da.$$

This expresses that f and $\hat{f}$ carry the same "energy".

4.3 Computing Fourier Transforms via Complex Analysis

Complex analysis, and in particular the residue theorem, provides a powerful method for computing Fourier transforms explicitly.

Example Let $f(x) = \dfrac{1}{1+x^2}$. Then

$$\hat{f}(a) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \frac{e^{-iax}}{1+x^2} \, dx.$$

We compute this using the residue theorem. Consider $a \leq 0$ and set

$$f(z) = \frac{e^{-iaz}}{1+z^2} = \frac{e^{-iaz}}{(z-i)(z+i)}.$$

Since $a \leq 0$, for $z = x + iy$ with $\mathrm{Im}(z) = y > 0$ we have $|e^{-iaz}| = e^{-a(-y)} = e^{ay} \cdot e^{???}$... more precisely, $e^{-iaz} = e^{-ia(x+iy)} = e^{-iax} e^{ay}$. Since $a \leq 0$ and $y > 0$, e^{ay} decays exponentially as $R \to \infty$. Hence the integral over the semicircle in the upper half-plane vanishes.

The only pole in the upper half-plane is at $z = i$. Using the residue theorem:

$$\int_{-\infty}^{\infty} f(x) \, dx = 2\pi i \; \mathrm{Res}_i(f).$$

Since i is a simple pole:

$$\mathrm{Res}_i(f) = \lim_{z \to i} (z-i) f(z) = \frac{e^{-ia \cdot i}}{2i} = \frac{e^a}{2i}.$$

Therefore

$$\int_{-\infty}^{\infty} \frac{e^{-iax}}{1+x^2} \, dx = 2\pi i \cdot \frac{e^a}{2i} = \pi e^a.$$

So for $a \leq 0$:

$$\mathcal{F}\left[\frac{1}{1+x^2}\right](a) = \frac{1}{\sqrt{2\pi}} \cdot \pi e^a = \sqrt{\frac{\pi}{2}} \, e^a.$$

A similar calculation for $a > 0$ using the semicircle in the lower half-plane gives $\sqrt{\frac{\pi}{2}} \, e^{-a}$. Combining both cases:

$$\boxed{\mathcal{F}\left[\frac{1}{1+x^2}\right](a) = \sqrt{\frac{\pi}{2}} \, e^{-|a|}.}$$

Chapter 5

Laplace Analysis

5.1 The Laplace Transform

Comparison with the Fourier transform:

- Fourier transform: computed for signals defined on $\mathbb{R}$.
- Laplace transform: computed for signals defined on $\mathbb{R}_+ = [0, \infty)$.

Both transforms convert differential equations into algebraic equations. The Laplace transform is a "generalization" in the sense that there are many functions which are not Fourier-transformable but are Laplace-transformable.

Definition 11 (Laplace Transform)**.** Let $f : \mathbb{R}_+ \to \mathbb{C}$. The **Laplace transform** of f at $z \in \mathbb{C}$, whenever well-defined, is

$$\mathcal{L}[f](z) := \int_0^\infty f(t)\, e^{-zt}\, dt.$$

Sometimes the Laplace transform of f is written by the corresponding capital letter, e.g. $F(z) = \mathcal{L}[f](z)$.

5.2 Convergence Criteria

Definition 12 (Abscissa of Convergence)**.** A value $\gamma_0 \in \mathbb{R}$ such that $\mathcal{L}[f](z)$ is well-defined at least on $\{\mathrm{Re}(z) > \gamma_0\}$ is called an **abscissa of convergence**.

The integral $\mathcal{L}[f](z)$ is well-defined if $\displaystyle\int_0^\infty |f(t)|\, e^{-z\,t}\, dt < \infty$.

Theorem 6 (Convergence Criterion)**.** *Assume there exists* $\gamma_0 \in \mathbb{R}$ *such that*

$$\int_0^\infty |f(t)|\, e^{-\gamma_0 t}\, dt < \infty$$

(intuitively, this means $|f(t)| < e^{\gamma_0 t}$ *as* $t \to \infty$*). Then for* $\mathrm{Re}(z) \geq \gamma_0$*:*

$$|\mathcal{L}[f](z)| = \int_0^\infty |f(t)|\, |e^{-\mathrm{Re}(z)\,t}||e^{-\mathrm{Im}(z)ti}|\, dt = \int_0^\infty |f(t)|\, e^{-\mathrm{Re}(z)\,t}\, dt \leq \int_0^\infty |f(t)|\, e^{-\gamma_0 t}\, dt < \infty.$$

Hence $\mathcal{L}[f](z)$ *is defined at least for* $\{z \in \mathbb{C} : \mathrm{Re}(z) > \gamma_0\}$*.*

Example If $f(t) = 0$ for $t \geq M$ and f is piecewise continuous, then

$$\int_0^\infty |f(t)|\, e^{-\gamma_0 t}\, dt = \int_0^M |f(t)|\, e^{-\gamma_0 t}\, dt < \infty$$

for all $\gamma_0 \in \mathbb{R}$. In this case $\mathcal{L}[f](z)$ is defined for all $z \in \mathbb{C}$.

Remark 10. Let γ_0 be an abscissa of convergence. Then $F(z) = \mathcal{L}[f](z)$ is **holomorphic** on $\{\mathrm{Re}(z) > \gamma_0\}$, and

$$F'(z) = \frac{d}{dz} \int_0^\infty f(t)\, e^{-zt}\, dt = \int_0^\infty f(t)(-t)\, e^{-zt}\, dt = -\mathcal{L}[tf(t)](z).$$

In short:

$$\boxed{\mathcal{L}[f(t)]' = -\mathcal{L}[tf(t)].}$$

(Here it is implicitly used that if $\gamma_0 \in \mathbb{R}$ is an abscissa of convergence for $\mathcal{L}[f(t)]$, it is also one for $\mathcal{L}[tf(t)]$.)

Example Let $f(t) = 1$. Then for every $\gamma_0 > 0$,

$$\int_0^\infty |f(t)|\, e^{-\gamma_0 t}\, dt = \int_0^\infty e^{-\gamma_0 t}\, dt = \frac{1}{\gamma_0} < \infty,$$

so any $\gamma_0 > 0$ is an abscissa of convergence, and $\mathcal{L}[f](z)$ is defined for $z \in \{\mathrm{Re}(z) > 0\}$. Moreover,

$$\mathcal{L}[1](z) = \int_0^\infty e^{-zt}\, dt = \left[\frac{e^{-zt}}{-z}\right]_0^\infty.$$

Since $\mathrm{Re}(z) > 0$ we have $\lim_{t\to\infty} e^{-zt} = 0$ (because $|e^{-zt}| = |e^{\mathrm{Re} z t}||e^{-\mathrm{Im} z t i}|$ (real part $\to 0$ and Im $= 1$ as $t \to \infty$)

$$\boxed{\mathcal{L}[1](z) = \frac{1}{z}, \quad \mathrm{Re}(z) > 0.}$$

Laplace transform of e^{at} For any $a \in \mathbb{R}$, $\gamma_0 = a$ is an abscissa of convergence for $\mathcal{L}[e^{at}](z)$, and

$$\boxed{\mathcal{L}[e^{at}](z) = \frac{1}{z-a}, \quad \mathrm{Re}(z) > a.}$$

5.3 Properties of the Laplace Transform

Example Let $f : \mathbb{R}_+ \to \mathbb{C}$, $f(t) = t$. By the derivative formula from the previous lecture, when $\mathrm{Re}(z) > 0$,

$$\frac{d}{dz}\mathcal{L}[1](z) = -\mathcal{L}[t](z) = -\mathcal{L}[f](z).$$

Since $\mathcal{L}[1](z) = \frac{1}{z}$, we get $\frac{d}{dz}\frac{1}{z} = -\frac{1}{z^2}$, hence

$$\boxed{\mathcal{L}[t](z) = \frac{1}{z^2}.}$$

Iterating this procedure gives the following.

Theorem 7. *For all $n \in \mathbb{N}$ and whenever* $\mathrm{Re}(z) > 0$,

$$\boxed{\mathcal{L}[t^n](z) = \frac{n!}{z^{n+1}}.}$$

Linearity Let $f, g : \mathbb{R}_+ \to \mathbb{C}$ be such that $\mathcal{L}[f]$ and $\mathcal{L}[g]$ are defined when $\mathrm{Re}(z) > \gamma_0$ for some $\gamma_0 \in \mathbb{R}$. Then for all $\lambda, \mu \in \mathbb{C}$, $\mathcal{L}[\lambda f + \mu g]$ is defined when $\mathrm{Re}(z) > \gamma_0$ and

$$\boxed{\mathcal{L}[\lambda f + \mu g](z) = \lambda\,\mathcal{L}[f](z) + \mu\,\mathcal{L}[g](z).}$$

Remark 11 (Nomenclature)**.** We say $f : \mathbb{R}_+ \to \mathbb{C}$ is of class C^k, $k \in \mathbb{N}$, if f is k-times differentiable and $f, f', f'', \dots, f^{(k)}$ are continuous (it suffices to require that $f^{(k)}$ is continuous).

5.3.1 Laplace Transform of Derivatives

Theorem 8. *Let $f : \mathbb{R}_+ \to \mathbb{C}$ be of class C^1. Assume $\gamma_0 \in \mathbb{R}$ is an abscissa of convergence for f and f'. Then for* $\mathrm{Re}(z) > \gamma_0$,

$$\boxed{\mathcal{L}[f'](z) = z\,\mathcal{L}[f](z) - f(0).}$$

Remark 12. Similarly to the Fourier transform, the Laplace transform converts derivatives into multiplication by z (hence differential equations into algebraic equations). A **crucial difference** is that the Laplace transform also picks up the “initial value” at 0.

Outline of the proof. By definition,

$$\mathcal{L}[f'](z) = \int_0^\infty f'(t)\,e^{-zt}\,dt.$$

We want to integrate by parts to remove the derivative from f. We have to be careful to treat the “boundary term” at ∞ appropriately. Since γ_0 is an abscissa of convergence for f' and $\mathrm{Re}(z) > \gamma_0$, the integral $\int_0^\infty f'(t)\,e^{-zt}\,dt$ converges absolutely. So for any sequence $T_n \to \infty$ as $n \to \infty$,

$$\mathcal{L}[f'](z) = \lim_{n\to\infty} \int_0^{T_n} f'(t)\,e^{-zt}\,dt.$$

Integration by parts gives

$$= \left[f(t)\,e^{-zt}\right]_0^{T_n} - \int_0^{T_n} f(t)(-z)\,e^{-zt}\,dt = \lim_{n\to\infty} f(T_n)\,e^{-T_n z} - f(0)\underbrace{e^{-0\cdot z}}_{=1} + z\underbrace{\int_0^\infty f(t)\,e^{-zt}\,dt}_{\text{converges absolutely}}.$$

Now choose T_n such that the boundary limit is 0. This is possible because, since γ_0 is an abscissa of convergence for f and $\mathrm{Re}(z) > \gamma_0$,

$$\int_0^\infty |f(t)|\,|e^{-zt}|\,dt \leq \int_0^\infty |f(t)|\,e^{-\mathrm{Re}(z)\,t}\,dt < \infty.$$

Thus there must exist a sequence $T_n \to \infty$ as $n \to \infty$ such that $|f(T_n)|\,|e^{-zT_n}| \to 0$ as $n \to \infty$ (otherwise $|f(t)|\,|e^{-zt}|$ would stay away from 0 as $t \to \infty$, contradicting the finiteness of $\int_0^\infty |f(t)|\,|e^{-zt}|\,dt$). When T_n is chosen that way, the above computations give

$$\mathcal{L}[f'](z) = -f(0) + z\,\mathcal{L}[f](z).$$ □

As a consequence of iterating this formula:

Corollary 5. *$f : \mathbb{R}_+ \to \mathbb{C}$ is of class C^k for $k \in \mathbb{N}$ and $\gamma_0 \in \mathbb{R}$ is an abscissa of convergence for $f, f', f'', \ldots, f^{(k)}$, then for $\mathrm{Re}(z) > \gamma_0$,*

$$\begin{aligned}
\mathcal{L}[f'](z) &= z\,\mathcal{L}[f](z) - f(0),\\
\mathcal{L}[f''](z) &= z\,\mathcal{L}[f'](z) - f'(0) = z\big[z\,\mathcal{L}[f](z) - f(0)\big] - f'(0),\\
&\ \vdots\\
\mathcal{L}[f^{(k)}](z) &= z^k\,\mathcal{L}[f](z) - \sum_{j=0}^{k-1} z^{k-1-j}\,f^{(j)}(0).
\end{aligned}$$

Example Consider $f(t) = t^2$, $f'(t) = 2t$, $f''(t) = 2$. Their Laplace transforms are defined when $\mathrm{Re}(z) > 0$:

$$\mathcal{L}[f](z) = \frac{2}{z^3}, \qquad \mathcal{L}[f'](z) = \frac{2}{z^2}, \qquad \mathcal{L}[f''](z) = \frac{2}{z}.$$

The relations $\mathcal{L}[f'] = z\,\mathcal{L}[f] - f(0)$ and $\mathcal{L}[f''] = z\,\mathcal{L}[f'] - f'(0)$ confirm our previous computations. Note carefully that we used $f(0) = f'(0) = 0$.

5.3.2 Laplace Transform of an Integral

Theorem 9. *Let $f : \mathbb{R}_+ \to \mathbb{C}$ be piecewise continuous and $\gamma_0 \geq 0$ an abscissa of convergence for f. Define $g : \mathbb{R}_+ \to \mathbb{C}$ by*

$$g(t) = \int_0^t f(s)\,ds.$$

Then when $\mathrm{Re}(z) > \gamma_0$,

$$\boxed{\mathcal{L}[g](z) = \frac{1}{z}\,\mathcal{L}[f](z).}$$

Remark 13. In general the abscissa of convergence for g cannot be negative (even if it is for f).

Outline. Note $g' = f$ and $g(0) = 0$. By the previous theorem,

$$\mathcal{L}[f](z) = \mathcal{L}[g'](z) = z\,\mathcal{L}[g](z) - \underbrace{g(0)}_{=0},$$

therefore $\mathcal{L}[g](z) = \frac{1}{z}\,\mathcal{L}[f](z)$. (We will not show why $\mathcal{L}[g](z)$ is indeed defined for $\mathrm{Re}(z) > \gamma_0$ when $\gamma_0 \geq 0$.) $\square$

5.3.3 Rescaling and Frequency Shift

Theorem 10. *Let $a > 0$ and $b \in \mathbb{R}$. Let $f : \mathbb{R}_+ \to \mathbb{C}$ be piecewise continuous with abscissa of convergence $\gamma_0 \in \mathbb{R}$. Define $g : \mathbb{R}_+ \to \mathbb{C}$ by*

$$g(t) = e^{-bt} f(at).$$

Then

$$\boxed{\mathcal{L}[g](z) = \frac{1}{a}\,\mathcal{L}[f]\left(\frac{z+b}{a}\right)}$$

holds when $\mathrm{Re}\left(\frac{z+b}{a}\right) > \gamma_0$ (equivalently, $\mathrm{Re}(z) > a\gamma_0 - b$, since $a > 0$ and $b \in \mathbb{R}$). In fact $a\gamma_0 - b$ is an abscissa of convergence for g (exercise).

Computation.

$$\mathcal{L}[g](z) = \int_0^\infty e^{-bt} f(at)\, e^{-zt}\, dt = \int_0^\infty f(at)\, e^{-(z+b)t}\, dt.$$

Substitute $s = at$, $ds = a\,dt$:

$$= \int_0^\infty f(s)\, e^{-(z+b)s/a}\, \frac{ds}{a} = \frac{1}{a}\int_0^\infty f(s)\, e^{-\frac{z+b}{a}\, s}\, ds = \frac{1}{a}\,\mathcal{L}[f]\left(\frac{z+b}{a}\right). \qquad \square$$

Remark 14. Compare this formula to the rescaling property of the Fourier transform.

Example Let $g(t) = e^{ct} = e^{ct} \cdot 1$. Here $a = 1$, $b = -c$. By the above theorem, when $\mathrm{Re}(z) > -b = c$,

$$\mathcal{L}[g](z) = \mathcal{L}[1](z-c) = \frac{1}{z-c}.$$

Remark 15. The fact that c is an abscissa of convergence for g can be shown by hand, without relying on the above theorem.

Example Find $f : \mathbb{R}_+ \to \mathbb{C}$ such that

$$\mathcal{L}[f](z) = \frac{1}{(z-a)(z-b)}, \qquad a, b \in \mathbb{R},\ a \neq b.$$

We compute

$$\mathcal{L}[f](z) = \frac{1}{b-a}\left[\frac{1}{z-b} - \frac{1}{z-a}\right] = \frac{1}{b-a}\left[\mathcal{L}[e^{bt}](z) - \mathcal{L}[e^{at}](z)\right] = \mathcal{L}\left[\frac{e^{bt} - e^{at}}{b-a}\right](z)$$

by linearity, defined for $\mathrm{Re}(z) > \max\{a, b\}$. Hence

$$f(t) = \frac{e^{bt} - e^{at}}{b-a}.$$

5.3.4 Convolution

Let $f, g : \mathbb{R}_+ \to \mathbb{C}$ be piecewise continuous. We extend these functions by 0 on $(-\infty, 0)$. Then

$$(f * g)(t) = \int_{-\infty}^{\infty} f(t-s)\, g(s)\, ds = \begin{cases} \displaystyle\int_0^t f(t-s)\, g(s)\, ds & \text{if } t > 0, \\ 0 & \text{otherwise.} \end{cases}$$

So in particular $f * g$ can be restricted back to $\mathbb{R}_+$, since its values on $(-\infty, 0)$ are trivial. That way we can regard $f * g$ as a function $\mathbb{R}_+ \to \mathbb{C}$ and compute its Laplace transform.

Theorem 11. *Let $f, g : \mathbb{R}_+ \to \mathbb{C}$ be piecewise continuous with common abscissa of convergence $\gamma_0 \in \mathbb{R}$. Then γ_0 is an abscissa of convergence for $f * g$ and when* $\mathrm{Re}(z) > \gamma_0$,

$$\boxed{\mathcal{L}[f * g](z) = \mathcal{L}[f](z)\, \mathcal{L}[g](z).}$$

Remark 16. As for the Fourier transform, the Laplace transform converts convolutions into products of Laplace transforms. Usually $\mathcal{L}[f]$ and $\mathcal{L}[g]$ are known; one then recovers $f * g$ as a "preimage".

Lecture 10 — 30-04-2026: Applications of Laplace Analysis to the Cauchy Problem

5.4 Applications of Laplace analysis to the Cauchy problem (initial value problem)

With the formulas from the previous lecture we can "invert" the Laplace transform of many functions. (There is a formal definition of the inverse Laplace transform that we will discover later.)

5.4.1 ODE with Constant Coefficients

First order initial value problem. Let $a, y_0 \in \mathbb{R}$ and $f : \mathbb{R}_+ \to \mathbb{R}$ piecewise continuous. **Task:** find a solution $y : \mathbb{R}_+ \to \mathbb{R}$ solving

$$(1) \qquad \begin{cases} y'(t) + a\, y(t) = f(t) & \text{for } t > 0, \\ y(0) = y_0 & \text{(initial condition).} \end{cases}$$

Existence: finding a solution (Laplace analysis). But before:

Uniqueness: there is a unique solution.

Remark 17. In general, linear initial value problems (i.e. no nonlinear dependence on the sought solution $y(t)$ such as $y(t)^2$, $y(t)^3$, etc.) have a unique solution if the number of initial conditions equals the **order** of the equation (highest appearing derivative).

Uniqueness for (1). Let $y_1, y_2 : \mathbb{R}_+ \to \mathbb{R}$ be two solutions of (1). **Strategy:** show they coincide; we'll do this by showing $w(t) = 0$ for all $t \in \mathbb{R}_+$, where $w(t) = y_1(t) - y_2(t)$.

We know y_1 and y_2 satisfy (1), so we want to derive a differential equation for w. Subtracting (1) for y_2 from (1) for y_1 yields

$$\begin{cases} w'(t) + a\,w(t) = 0, \\ w(0) = 0. \end{cases}$$

Multiply w by e^{at}. Then the previous equation gives

$$\left(e^{at}w(t)\right)' = a\,e^{at}w(t) + e^{at}w'(t) = a\,e^{at}w(t) - a\,e^{at}w(t) = 0$$

(using the equation for w). Hence $e^{at}w(t)$ is constant in $t \in \mathbb{R}_+$. Therefore

$$e^{at}w(t) = e^{a\cdot 0}w(0) = 0 \quad \Longrightarrow \quad w(t) = 0 \text{ for all } t \in \mathbb{R}_+.$$

Existence. Find a solution by taking the Laplace transform of (1):

$$y'(t) + a\,y(t) = f(t)$$

Apply Laplace (formally, assuming "everything is well-defined"):

$$(2) \qquad \mathcal{L}[y'](z) + a\,\mathcal{L}[y](z) = \mathcal{L}[f](z).$$

Setting $Y = \mathcal{L}[y]$ and $F = \mathcal{L}[f]$ and using the formula for the Laplace transform of derivatives,

$$(2) \implies z\,Y(z) - \underbrace{y(0)}_{=y_0} + a\,Y(z) = F(z).$$

Isolating Y:

$$Y(z) = \frac{F(z)}{z+a} + \frac{y_0}{z+a}.$$

Punchline: derivatives transform into algebraic equations!

Is the right-hand side of the above expression for Y the Laplace transform of something? Recall $\frac{1}{z+a} = \mathcal{L}[e^{-at}](z)$. Hence

$$Y(z) = \underbrace{\mathcal{L}[e^{-at}](z)\,\mathcal{L}[f](z)}_{\substack{\text{Laplace transform of convolution} \\ \text{of } e^{-at} \text{ with } f(t)}} + y_0\,\mathcal{L}[e^{-at}](z).$$

Formally inverting both sides:

$$\boxed{y(t) = \int_0^t e^{-a(t-s)} f(s)\,ds + y_0\,e^{-at}.}$$

Indeed, one can verify this is a solution of (1).

5.4.2 Second Order Initial Value Problem

Let $\omega \in \mathbb{R}_+$, $a > 0$, and $f : \mathbb{R}_+ \to \mathbb{R}$ as above. Consider

$$(3) \qquad \begin{cases} y''(t) + 2a\,y'(t) + \omega^2\,y(t) = f(t) & \text{for } t > 0, \\ y(0) = y_0, \\ y'(0) = v_0. \end{cases}$$

Remark 18. This is the equation satisfied by the **damped harmonic oscillator**: f is the forcing, a is the damping parameter coming from friction.

Uniqueness. Let $y_1, y_2 : \mathbb{R}_+ \to \mathbb{R}$ be two solutions of (3). Set $u(t) = y_1(t) - y_2(t)$. As above we want to show $u(t) = 0$ for all $t \in \mathbb{R}_+$, this time by using the **energy method**.
Subtracting (3) for y_2 from (3) for y_1:

$$\begin{cases} u''(t) + 2a\,u'(t) + \omega^2\,u(t) = 0, \\ u(0) = 0, \\ u'(0) = 0. \end{cases}$$

Multiply the equation by u':

$$\underbrace{u''\,u'}_{=\frac{1}{2}((u')^2)'} + 2a(u')^2 + \omega^2 \underbrace{u'\,u}_{=\frac{1}{2}(u^2)'} = 0$$

$$\Longrightarrow \left(\tfrac{1}{2}(u')^2\right)' + 2a(u')^2 + \left(\tfrac{1}{2}\omega^2 u^2\right)' = 0$$

$$\Longrightarrow \left(\underbrace{\tfrac{1}{2}(u')^2}_{\text{kinetic energy}} + \underbrace{\tfrac{1}{2}\omega^2 u^2}_{\text{potential energy}}\right)' = -2a(u')^2 \le 0.$$

This means the energy

$$E(t) = \tfrac{1}{2}u'(t)^2 + \tfrac{1}{2}\omega^2 u(t)^2$$

is **nonincreasing** (effect of damping). Since

- $E(0) = 0$ (because $u(0) = 0$ and $u'(0) = 0$),
- $E(t) \ge 0$ (since it's a sum of squares),

we get $E(t) = 0$ for all $t \in \mathbb{R}_+$, hence $u(t) = 0$ for all $t \in \mathbb{R}_+$, since

$$0 \le \tfrac{1}{2}\omega^2 u^2 \le \tfrac{1}{2}\omega^2 u^2 + \tfrac{1}{2}(u')^2 = E = 0.$$

Both sides of this chain are 0, which forces $\frac{1}{2}\omega^2 u^2 = 0$ and thus $u^2 = 0$, as desired.

Existence. By applying Laplace transform to the equation $y''(t) + 2a\,y'(t) + \omega^2\,y(t) = f(t)$ and using the identities for derivatives:

$$z^2 Y(z) - z\,\underbrace{y(0)}_{=y_0} - \underbrace{y'(0)}_{=v_0} + 2a\big[z\,Y(z) - y_0\big] + \omega^2 Y(z) = F(z).$$

Rearranging:

$$(z^2 + 2az + \omega^2)\,Y(z) = F(z) + (z+a)\,y_0 + a\,y_0 + v_0.$$

Completing the square: $z^2 + 2az + \omega^2 = (z+a)^2 + \omega^2 - a^2$. Hence

$$Y(z) = \frac{F(z)}{(z+a)^2 + \omega^2 - a^2} + \frac{z+a}{(z+a)^2 + \omega^2 - a^2}\,y_0 + \frac{1}{(z+a)^2 + \omega^2 - a^2}\,(ay_0 + v_0).$$

We now consider three cases depending on the sign of $\omega^2 - a^2$.

Case (1): $a^2 = \omega^2$ **(critically damped)**

In this case
$$\frac{1}{(z+a)^2+\omega^2-a^2} = \frac{1}{(z+a)^2}, \qquad \frac{z+a}{(z+a)^2+\omega^2-a^2} = \frac{1}{z+a}.$$
Recall $\mathcal{L}[e^{-at}](z) = \frac{1}{z+a}$ and
$$\frac{1}{(z+a)^2} = -\frac{d}{dz}\frac{1}{z+a} = -\frac{d}{dz}\mathcal{L}[e^{-at}](z) = \mathcal{L}[t\,e^{-at}](z).$$
Hence
$$Y(z) = \mathcal{L}[t\,e^{-at}](z)\,\mathcal{L}[f](z) + \mathcal{L}[e^{-at}](z)\,y_0 + \mathcal{L}[t\,e^{-at}](z)\,(ay_0+v_0),$$
and formally inverting,
$$\boxed{y(t) = \int_0^t (t-s)\,e^{-a(t-s)}f(s)\,ds + e^{-at}\,y_0 + t\,e^{-at}\,(ay_0+v_0).}$$

Case (2): $\omega^2 > a^2$ **(underdamped)**

We start from
$$\boxed{\mathcal{L}[\cos(\beta t)](z) = \frac{z}{z^2+\beta^2}, \qquad \mathcal{L}[\sin(\beta t)](z) = \frac{\beta}{z^2+\beta^2}.}$$
(*Exercise:* check this is consistent with $\mathcal{L}[e^{-i\beta t}](z) = \frac{1}{z+i\beta}$.)

Let $\beta = \sqrt{\omega^2-a^2}$. Again rewrite the appearing reciprocals:
$$\frac{1}{(z+a)^2+\underbrace{\omega^2-a^2}_{=\beta^2}} = \frac{1}{\beta}\cdot\frac{\beta}{(z+a)^2+\beta^2} = \frac{1}{\beta}\,\mathcal{L}[e^{-at}\sin(\beta t)](z) \qquad \text{(frequency shift)},$$
$$\frac{z+a}{(z+a)^2+\underbrace{\omega^2-a^2}_{=\beta^2}} = \mathcal{L}[e^{-at}\cos(\beta t)](z).$$
Hence
$$Y(z) = \mathcal{L}\left[\frac{1}{\beta}\,e^{-at}\sin(\beta t)\right](z)\,\mathcal{L}[f](z) + \mathcal{L}[e^{-at}\cos(\beta t)](z)\,y_0 + \mathcal{L}\left[\frac{1}{\beta}\,e^{-at}\sin(\beta t)\right](z)\,(ay_0+v_0).$$
Therefore
$$\boxed{\begin{aligned} y(t) = &\int_0^t \frac{1}{\sqrt{\omega^2-a^2}}\,e^{-a(t-s)}\sin\Big(\sqrt{\omega^2-a^2}\,(t-s)\Big)\,f(s)\,ds \\ &+ y_0\,e^{-at}\cos\Big(\sqrt{\omega^2-a^2}\,t\Big) \\ &+ (ay_0+v_0)\,\frac{1}{\sqrt{\omega^2-a^2}}\,e^{-at}\sin\Big(\sqrt{\omega^2-a^2}\,t\Big). \end{aligned}}$$

Case (3): $\omega^2 < a^2$ **(overdamped)**

Recall

$$\boxed{\mathcal{L}[\cosh(\beta t)](z) = \frac{z}{z^2 - \beta^2}, \qquad \mathcal{L}[\sinh(\beta t)](z) = \frac{\beta}{z^2 - \beta^2}.}$$

(*Exercise:* this can be derived from the formulas in case (2) using $\cosh(x) = \cos(ix)$ and $\sinh(x) = -i\sin(ix)$.)
Let $\beta = \sqrt{a^2 - \omega^2}$. Then

$$\frac{1}{(z+a)^2 + \underbrace{\omega^2 - a^2}_{=-\beta^2}} = \frac{1}{\beta} \cdot \frac{\beta}{(z+a)^2 - \beta^2} = \frac{1}{\beta}\,\mathcal{L}[e^{-at}\sinh(\beta t)](z),$$

$$\frac{z+a}{(z+a)^2 + \underbrace{\omega^2 - a^2}_{=-\beta^2}} = \mathcal{L}[e^{-at}\cosh(\beta t)](z).$$

Hence

$$\begin{aligned} Y(z) = {} & \mathcal{L}\left[\frac{1}{\sqrt{a^2-\omega^2}}\, e^{-at}\sinh\Big(\sqrt{a^2-\omega^2}\,t\Big)\right](z)\,\mathcal{L}[f](z) + \mathcal{L}[e^{-at}\cosh(\sqrt{a^2-\omega^2}\,t)](z)\,y_0 \\ & + \mathcal{L}\left[\frac{1}{\sqrt{a^2-\omega^2}}\, e^{-at}\sinh\Big(\sqrt{a^2-\omega^2}\,t\Big)\right](z)\,(ay_0 + v_0). \end{aligned}$$

Therefore

$$\boxed{\begin{aligned} y(t) = {} & \int_0^t \frac{1}{\sqrt{a^2-\omega^2}}\, e^{-a(t-s)}\sinh\Big(\sqrt{a^2-\omega^2}\,(t-s)\Big)\, f(s)\,ds \\ & + y_0\, e^{-at}\cosh\Big(\sqrt{a^2-\omega^2}\,t\Big) \\ & + (ay_0 + v_0)\,\frac{1}{\sqrt{a^2-\omega^2}}\, e^{-at}\sinh\Big(\sqrt{a^2-\omega^2}\,t\Big). \end{aligned}}$$

5.4.3 General Strategy to Find a Solution

1. Apply Laplace transform to the equation.
2. Using calculus rules, isolate $\mathcal{L}[y]$.
3. "Detect" known Laplace transforms on the right-hand side.
4. "Invert": $\mathcal{L}[y] = \mathcal{L}[\text{detected function}]$.

5.5 Inverse Laplace Transform

Given $\mathcal{L}[f]$, can we determine f in general?

Recall: if $\gamma_0 \in \mathbb{R}$ is an abscissa of convergence for f, then $\mathcal{L}[f]$ is holomorphic (hence without singularities) on $\{\mathrm{Re}(z) > \gamma_0\}$. In fact, the "opposite" is also true: if $\mathcal{L}[f]$ is holomorphic on $\{\mathrm{Re}(z) > \gamma_0\}$ and "doesn't grow" as $|z| \to \infty$, then γ_0 is an abscissa of convergence for f.

Theorem 12 (Inverse Laplace transform)**.** *Assume $f : \mathbb{R}_+ \to \mathbb{C}$ is piecewise continuous, and for some $\gamma \in \mathbb{R}$, $\mathcal{L}[f]$ is holomorphic on $\{\mathrm{Re}(z) \geq \gamma\}$ and*

$$\lim_{T\to\infty} \int_{-T}^{T} F(\gamma + is)\, e^{(\gamma+is)t}\, ds$$

exists for all $t \in \mathbb{R}_+$. (Both hold whenever there exists $\gamma_0 < \gamma$ abscissa of convergence.) Then

$$\boxed{f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \mathcal{L}[f](\gamma + is)\, e^{(\gamma+is)t}\, ds.}$$

Remark 19. The integral on the right-hand side need not converge absolutely, so it is understood as

$$\lim_{T\to\infty} \int_{-T}^{T} F(\gamma + is)\, e^{(\gamma+is)t}\, ds.$$

How to show this formula? (Outlook)

Extend $f : \mathbb{R}_+ \to \mathbb{C}$ to a function defined on $\mathbb{R}$ by setting it to be 0 on $(-\infty, 0)$. Let $g(t) = f(t)\, e^{-\gamma t}$. Take the Fourier transform:

$$\hat{g}(a) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} g(t)\, e^{-iat}\, dt = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(t)\, e^{-(\gamma+ia)t}\, dt.$$

Since $f = 0$ on $(-\infty, 0)$,

$$\hat{g}(a) = \frac{1}{\sqrt{2\pi}} \int_{0}^{\infty} f(t)\, e^{-(\gamma+ia)t}\, dt = \mathcal{L}[f](\gamma + ia).$$

Taking the inverse Fourier transform,

$$g(t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \hat{g}(a)\, e^{iat}\, da = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\gamma + ia)\, e^{iat}\, da.$$

Therefore

$$f(t) = e^{\gamma t}\, g(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\gamma + ia)\, e^{(\gamma+ia)t}\, da.$$

Remark 20.
1. **Lerch's theorem:** f is uniquely determined by $\mathcal{L}[f]$ through this formula.
2. The condition $\lim_{T\to\infty} \int_{-T}^{T} F(\gamma+is)\, e^{(\gamma+is)t}\, ds < \infty$ is automatically true provided $\int_{-\infty}^{\infty} |F(\gamma+is)|\, ds < \infty$.

5.5.1 Examples of Inversion

Example Let

$$F(z) = \frac{1}{(z+1)(z+2)^2}.$$

We want to find $f : \mathbb{R}_+ \to \mathbb{C}$ such that $\mathcal{L}[f] = F$. Two ways to do this.

(1) Partial fraction decomposition.

$$\frac{1}{(z+1)(z+2)^2} = \underbrace{\frac{1}{z+1}}_{=\mathcal{L}[e^{-t}]} - \underbrace{\frac{1}{z+2}}_{=\mathcal{L}[e^{-2t}]} - \underbrace{\frac{1}{(z+2)^2}}_{=\mathcal{L}[t\,e^{-2t}]}.$$

So

$$f(t) = e^{-t} - e^{-2t} - t\,e^{-2t}.$$

By **uniqueness** of the Laplace transform, this is enough. Usually this is the fastest way.

(2) Inverse Laplace transform and residue theorem.

$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\gamma + is)\, e^{(\gamma+is)t}\, ds.$$

Let $h(z) = F(z)\, e^{zt}$. The poles of h are at $z = -1$ and $z = -2$. We choose $\gamma = 0$:

- F is holomorphic on $\{\mathrm{Re}(z) > -1\}$;
- $\int_{-\infty}^{\infty} |F(\gamma + is)|\, ds < \infty$.

The inverse Laplace transform formula reads

$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} h(\gamma + is)\, ds = \frac{1}{2\pi i} \int_{-\infty i}^{\infty i} h(z)\, dz.$$

Define the curves

$$L_r : [-r, r] \to \mathbb{C}, \quad s \mapsto si, \qquad C_r : \left[\tfrac{\pi}{2}, \tfrac{3\pi}{2}\right] \to \mathbb{C}, \quad \vartheta \mapsto r\, e^{i\vartheta},$$

and let $\Gamma = L_r \cup C_r$ oriented counterclockwise (so C_r closes the contour through the *left* half-plane). When $r > 2$, Γ encloses all poles of h, namely -1 and -2. By the residue theorem,

$$\int_\Gamma h(z)\, dz = 2\pi i \big[\mathrm{Res}_{-1}(h) + \mathrm{Res}_{-2}(h)\big].$$

Compute the residues. Looking at h we find:

- $z = -1$ simple pole:

$$\mathrm{Res}_{-1}(h) = \lim_{z \to -1} \big((z+1)\, h(z)\big) = \lim_{z \to -1} \frac{e^{zt}}{(z+2)^2} = e^{-t}.$$

- $z = -2$ pole of order 2:

$$\mathrm{Res}_{-2}(h) = \lim_{z \to -2} \frac{d}{dz} \big((z+2)^2\, h(z)\big) = \lim_{z \to -2} \frac{d}{dz} \left(\frac{e^{zt}}{z+1}\right) = -e^{-2t}(t+1).$$

Hence

$$\int_\Gamma h(z)\, dz = 2\pi i \big[e^{-t} - e^{-2t} - t\, e^{-2t}\big].$$

Now

$$\frac{1}{2\pi i} \int_\Gamma h(z)\, dz = \underbrace{\frac{1}{2\pi i} \int_{L_r} h(z)\, dz}_{\to f(t) \text{ as } r \to \infty} + \frac{1}{2\pi i} \int_{C_r} h(z)\, dz,$$

and the left-hand side equals $e^t - e^{-2t} - t\,e^{-2t}$.

Now we want to show $\left|\int_{C_r} h(z)\,dz\right| \to 0$ **as** $r \to \infty$**.**

$$\left|\int_{C_r} h(z)\,dz\right| = \left|\int_{\pi/2}^{3\pi/2} h(r\,e^{i\vartheta})\,i\,r\,e^{i\vartheta}\,d\vartheta\right| \overset{\Delta\text{-ineq.}}{\leq} \int_{\pi/2}^{3\pi/2} \left|h(r\,e^{i\vartheta})\right|\,r\,d\vartheta$$

(using $|i\,r\,e^{i\vartheta}| = r$)

$$= \int_{\pi/2}^{3\pi/2} \frac{\left|e^{r\,e^{i\vartheta}\,t}\right|}{\left|r\,e^{i\vartheta}+1\right|\,\left|r\,e^{i\vartheta}+2\right|^2}\,r\,d\vartheta,$$

where the denominators are of order $\sim r$ and $\sim r^2$ respectively.

Notice:

$$\left|e^{r\,e^{i\vartheta}\,t}\right| = \left|e^{r(\cos\vartheta + i\sin\vartheta)t}\right| = \left|e^{r\cos\vartheta\,t}\right|\,\underbrace{\left|e^{ir\sin\vartheta\,t}\right|}_{=1} = e^{r\cos\vartheta\,t} \leq 1$$

because $\cos\vartheta \leq 0$ for $\vartheta \in [\frac{\pi}{2}, \frac{3\pi}{2}]$.

Hence $\left|\int_{C_r} h(z)\,dz\right|$ can be bounded from above by a term of order

$$\int_{\pi/2}^{3\pi/2} \frac{1}{r\cdot r^2}\,r\,d\vartheta \to 0 \quad \text{as } r \to \infty.$$

Remark 21. We close the **left half-circle** (for $\mathrm{Re}(z) < \gamma$) because in that region $|e^{zt}| \leq e^{\gamma t}$. This guarantees the control on the exponential term above.

5.6 Applications to Partial Differential Equations (PDEs)

As for ODEs, we will apply the theory of Laplace and Fourier transforms to solve initial-boundary value problems for **linear** PDEs with **constant coefficients**, e.g. the **heat equation**:

$$\begin{cases} \dfrac{\partial u}{\partial t}(x,t) - \dfrac{\partial^2 u}{\partial x^2}(x,t) = f(x,t) & \text{for } t > 0 \text{ and } x \in (0,L), \\ u(x,0) = u_0(x) & \text{(initial condition)}, \\ u(0,t) = 0,\ u(L,t) = 0 & \text{(boundary condition)}. \end{cases}$$

This describes the evolution of temperature u of a one-dimensional rod of length L with heat source f and fixed temperature at the endpoints 0 and L.

General procedure that we will follow.

- **Fourier series** if space or time domain is bounded.
- **Fourier transform** if space or time domain is $\mathbb{R}$.
- **Laplace transform** if space or time domain is $\mathbb{R}_+$.

5.6.1 Fourier Series Recapitulation

We want to decompose $f : [0, L] \to \mathbb{R}$ as an infinite sum of oscillating functions. **Basis functions:**

- $\sin\left(\dfrac{2\pi n}{L}\,x\right)$ where $n \geq 1$,
- $\cos\left(\dfrac{2\pi n}{L}\,x\right)$ where $n \geq 0$.

Orthogonality. With the inner product

$$\langle f, g\rangle := \int_0^L f(x)\,g(x)\,dx,$$

one has:

- $\left\langle \sin\left(\frac{2\pi n}{L}\,x\right), \sin\left(\frac{2\pi m}{L}\,x\right)\right\rangle = \begin{cases} L/2 & \text{if } n = m, \\ 0 & \text{otherwise.} \end{cases}$
- Same for $\left\langle \cos\left(\frac{2\pi n}{L}\,x\right), \cos\left(\frac{2\pi m}{L}\,x\right)\right\rangle$.
- $\left\langle \sin\left(\frac{2\pi n}{L}\,x\right), \cos\left(\frac{2\pi m}{L}\,x\right)\right\rangle = 0$ for all n, m.

We will use these facts as an orthogonal basis to decompose other functions via **trigonometric (Fourier) series**.
Given $f : [0, L] \to \mathbb{R}$ piecewise continuous, can we write

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty}\left[a_n \cos\left(\frac{2\pi n}{L}\,x\right) + b_n \sin\left(\frac{2\pi n}{L}\,x\right)\right]?$$

(The same considerations apply when $f : \mathbb{R} \to \mathbb{R}$ is L-periodic, hence can be thought of as a function on $[0, L]$.)
The answer is **yes**!

Computing the coefficients. To "guess" the form of the coefficients a_n and b_n, we assume f already has this form. Then we can (formally) compute the following using orthogonality:

- $\displaystyle\int_0^L f(x)\,dx = \frac{L}{2}\,a_0,$
- $\displaystyle\int_0^L f(x)\cos\left(\frac{2\pi n}{L}\,x\right)dx = \frac{L}{2}\,a_n$ where $n \geq 1$,
- $\displaystyle\int_0^L f(x)\sin\left(\frac{2\pi n}{L}\,x\right)dx = \frac{L}{2}\,b_n$ where $n \geq 1$.

This suggests defining

$$\boxed{a_n = \frac{2}{L}\int_0^L f(x)\cos\left(\frac{2\pi n}{L}\,x\right)dx \quad \text{where } n \geq 0,}$$

$$\boxed{b_n = \frac{2}{L}\int_0^L f(x)\sin\left(\frac{2\pi n}{L}\,x\right)dx \quad \text{where } n \geq 1.}$$

Trigonometric Fourier decomposition. Define $f_N : [0, L] \to \mathbb{R}$ by

$$f_N(x) = \frac{a_0}{2} + \sum_{n=1}^{N} \left[a_n \cos\left(\frac{2\pi n}{L}\,x\right) + b_n \sin\left(\frac{2\pi n}{L}\,x\right) \right],$$

where a_n and b_n are as above and $f : [0, L] \to \mathbb{R}$ is piecewise continuous. Then:

- $f_N(x) \to f(x)$ as $N \to \infty$ at every $x \in [0, L]$ where f is differentiable;
- $f_N(x) \to \dfrac{f(x^+) + f(x^-)}{2}$ when x is a jump discontinuity of f;
- $\displaystyle\lim_{N\to\infty} \int_0^L |f_N(x) - f(x)|\,dx = 0.$

5.6.2 Special Cases: Even and Odd Functions

Odd functions. Assume f is **odd** (i.e. $f(-x) = -f(x)$ for all $x \in [0, L]$). Then $a_n = 0$ for every $n \geq 0$ and

$$f(x) = \sum_{n=1}^{\infty} b_n \sin\left(\frac{2\pi n}{L}\,x\right),$$

where

$$b_n = \frac{2}{L}\int_0^L f(x) \sin\left(\frac{2\pi n}{L}\,x\right) dx = \frac{2}{L}\int_{-L/2}^{L/2} f(x) \sin\left(\frac{2\pi n}{L}\,x\right) dx$$

(shift in domain allowed since all functions involved are L-periodic), so

$$b_n = \frac{4}{L}\int_0^{L/2} f(x) \sin\left(\frac{2\pi n}{L}\,x\right) dx$$

since f and $\sin\left(\frac{2\pi n}{L}\,x\right)$ are both odd, so their product is even.

Even functions. Assume f is **even** (i.e. $f(-x) = f(x)$ for all $x \in [0, L]$). Then $b_n = 0$ for all $n \geq 1$ and

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{2\pi n}{L}\,x\right),$$

where

$$a_n = \frac{2}{L}\int_0^L f(x) \cos\left(\frac{2\pi n}{L}\,x\right) dx = \frac{2}{L}\int_{-L/2}^{L/2} f(x) \cos\left(\frac{2\pi n}{L}\,x\right) dx$$

by L-periodicity, so

$$a_n = \frac{4}{L}\int_0^{L/2} f(x) \cos\left(\frac{2\pi n}{L}\,x\right) dx$$

since f and $\cos\left(\frac{2\pi n}{L}\,x\right)$ are both even, so their product is even.

Remark 22.

- The functions $a_n \cos\left(\frac{2\pi n}{L}\,x\right)$ and $b_n \sin\left(\frac{2\pi n}{L}\,x\right)$ are sometimes called **modes**.
- The function $e_n(x) = \cos\left(\frac{2\pi n}{L}\,x\right)$ satisfies **Neumann boundary conditions**:

$$\boxed{e_n'(0) = 0, \qquad e_n'(L) = 0.}$$

- The function $h_n(x) = \sin\left(\frac{2\pi n}{L}\,x\right)$ satisfies **Dirichlet boundary conditions**:

$$\boxed{h_n(0) = 0, \qquad h_n(L) = 0.}$$